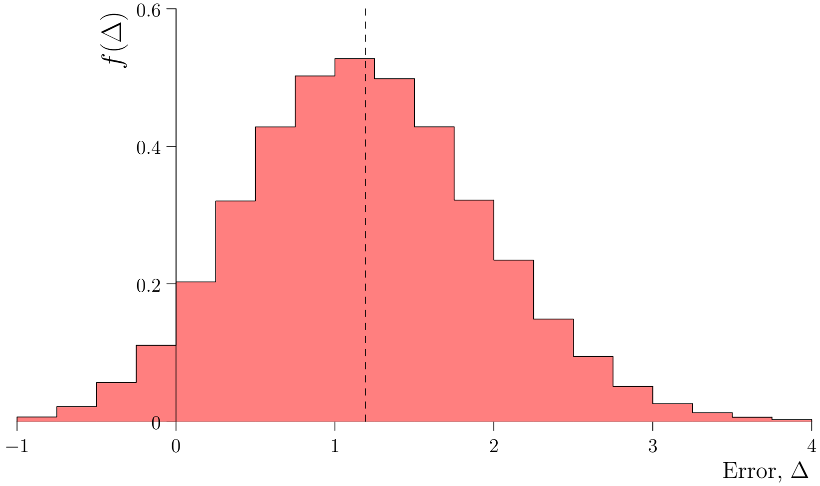
DISC-NET ML Workshop

Lessons from Machine Learning Mistakes

Data sets, reproducibility and metrics

Outline

1. **Data Sets**
2. Reproducibility
3. Metrics



History of Doing Things Badly

- We learn from out mistakes
- The history of machine learning is full of mistakes
- That is, doing thing poorly leading to misunderstanding and confusion in the literature

History of Doing Things Badly

- We learn from out mistakes
- The history of machine learning is full of mistakes
- That is, doing thing poorly leading to misunderstanding and confusion in the literature

History of Doing Things Badly

- We learn from out mistakes
- The history of machine learning is full of mistakes
- That is, doing thing poorly leading to misunderstanding and confusion in the literature

History of Doing Things Badly

- We learn from out mistakes
- The history of machine learning is full of mistakes
- That is, doing thing poorly leading to misunderstanding and confusion in the literature
- This still happens

History of Doing Things Badly

- We learn from out mistakes
- The history of machine learning is full of mistakes
- That is, doing thing poorly leading to misunderstanding and confusion in the literature
- This still happens, all the time

History of Doing Things Badly

- We learn from out mistakes
- The history of machine learning is full of mistakes
- That is, doing thing poorly leading to misunderstanding and confusion in the literature
- This still happens, all the time, but we have learned to do things better

Datasets

- Machine Learning is driven by the availability of datasets
- In deep learning these have to be large
- Of course, to be useful to commoners like you and me, we also want them to be easy to handle

Datasets

- Machine Learning is driven by the availability of datasets
- In deep learning these have to be large
- Of course, to be useful to commoners like you and me, we also want them to be easy to handle

Datasets

- Machine Learning is driven by the availability of datasets
- In deep learning these have to be large, and initially labelled
- Of course, to be useful to commoners like you and me, we also want them to be easy to handle

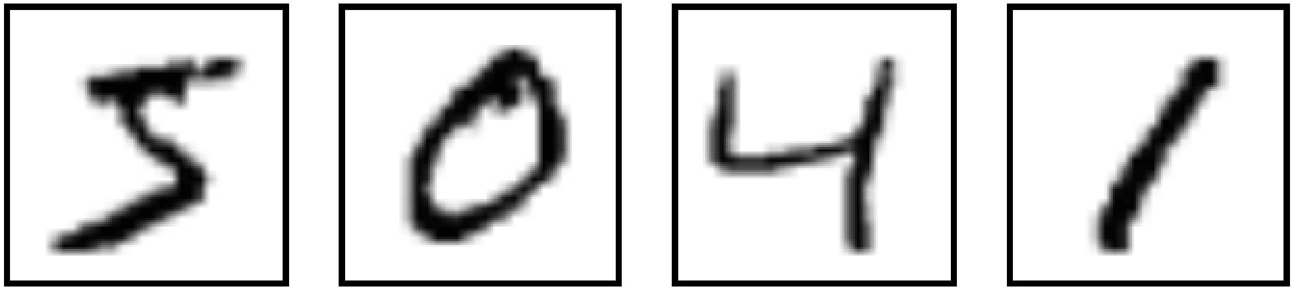
Datasets

- Machine Learning is driven by the availability of datasets
- In deep learning these have to be large, and initially labelled
- Of course, to be useful to commoners like you and me, we also want them to be easy to handle

Datasets

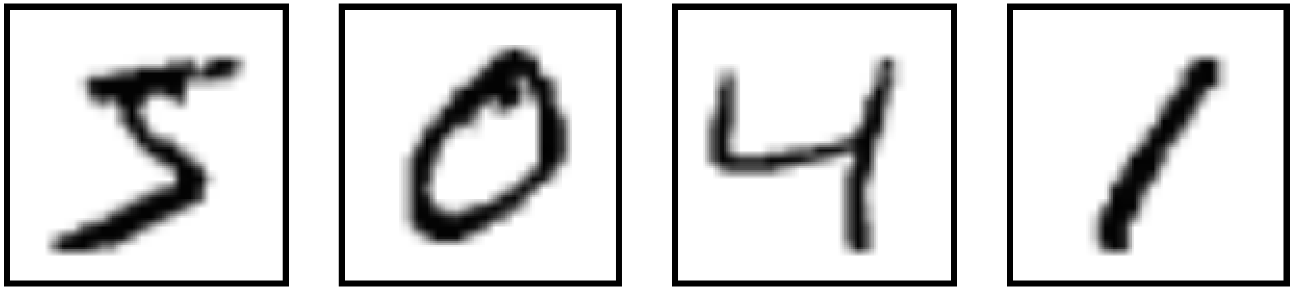
- Machine Learning is driven by the availability of datasets
- In deep learning these have to be large, and initially labelled
- Of course, to be useful to commoners like you and me, we also want them to be easy to handle (small as we can get away with)

MNIST



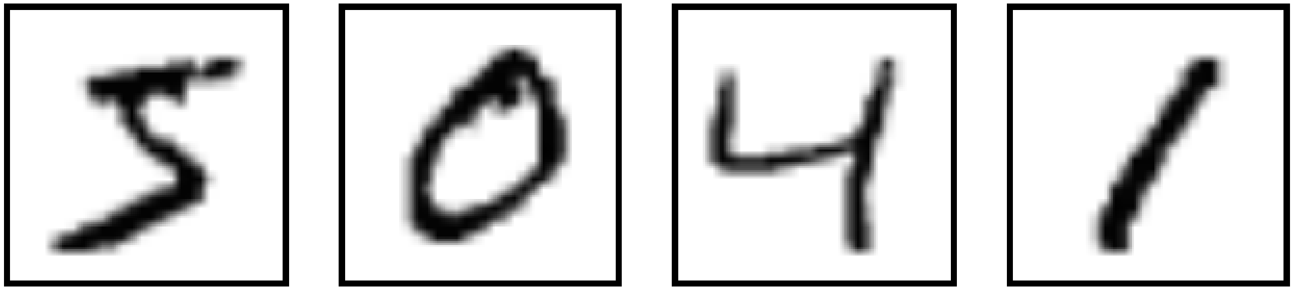
- 28×28 black and white hand-written digits
- It had 60 000 training images (roughly 6000 per digit)
- 10 000 test images
- It is really fast to work with

MNIST



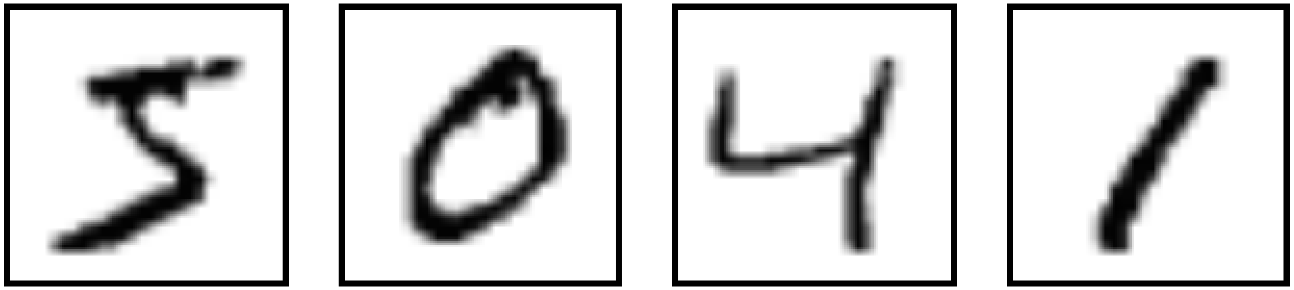
- 28×28 black and white hand-written digits
- It had 60 000 training images (roughly 6000 per digit)
- 10 000 test images
- It is really fast to work with

MNIST



- 28×28 black and white hand-written digits
- It had 60 000 training images (roughly 6000 per digit)
- 10 000 test images
- It is really fast to work with

MNIST



- 28×28 black and white hand-written digits
- It had 60 000 training images (roughly 6000 per digit)
- 10 000 test images
- It is really fast to work with

Sins of MNIST

- MNIST isn't that hard
- The digits are already centred and rescaled so it is not that interesting
- Many practitioner's would run test multiple times and state their best performance
- For most good networks the task is so easy that the errors are where the scaling and centring went wrong or the digits are unintelligible
- Ignoring this practitioners would start over-fitting on the test set

Sins of MNIST

- MNIST isn't that hard (MLPs can do a reasonable job)
- The digits are already centred and rescaled so it is not that interesting
- Many practitioner's would run test multiple times and state their best performance
- For most good networks the task is so easy that the errors are where the scaling and centring went wrong or the digits are unintelligible
- Ignoring this practitioners would start over-fitting on the test set

Sins of MNIST

- MNIST isn't that hard (MLPs can do a reasonable job)
- The digits are already centred and rescaled so it is not that interesting
- Many practitioner's would run test multiple times and state their best performance
- For most good networks the task is so easy that the errors are where the scaling and centring went wrong or the digits are unintelligible
- Ignoring this practitioners would start over-fitting on the test set

Sins of MNIST

- MNIST isn't that hard (MLPs can do a reasonable job)
- The digits are already centred and rescaled so it is not that interesting
- Many practitioner's would run test multiple times and state their best performance
- For most good networks the task is so easy that the errors are where the scaling and centring went wrong or the digits are unintelligible
- Ignoring this practitioners would start over-fitting on the test set

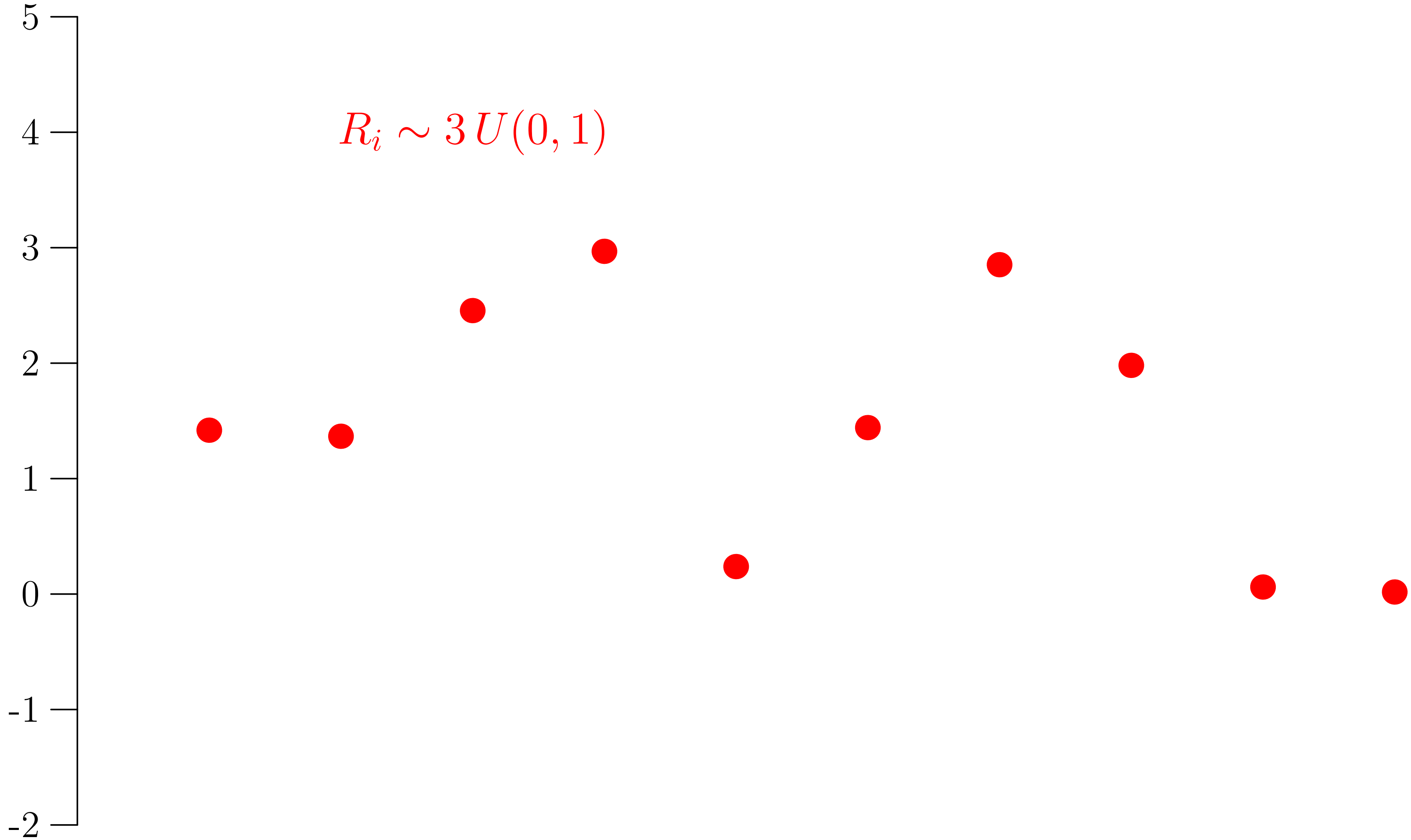
Sins of MNIST

- MNIST isn't that hard (MLPs can do a reasonable job)
- The digits are already centred and rescaled so it is not that interesting
- Many practitioner's would run test multiple times and state their best performance
- For most good networks the task is so easy that the errors are where the scaling and centring went wrong or the digits are unintelligible
- Ignoring this practitioners would start over-fitting on the test set

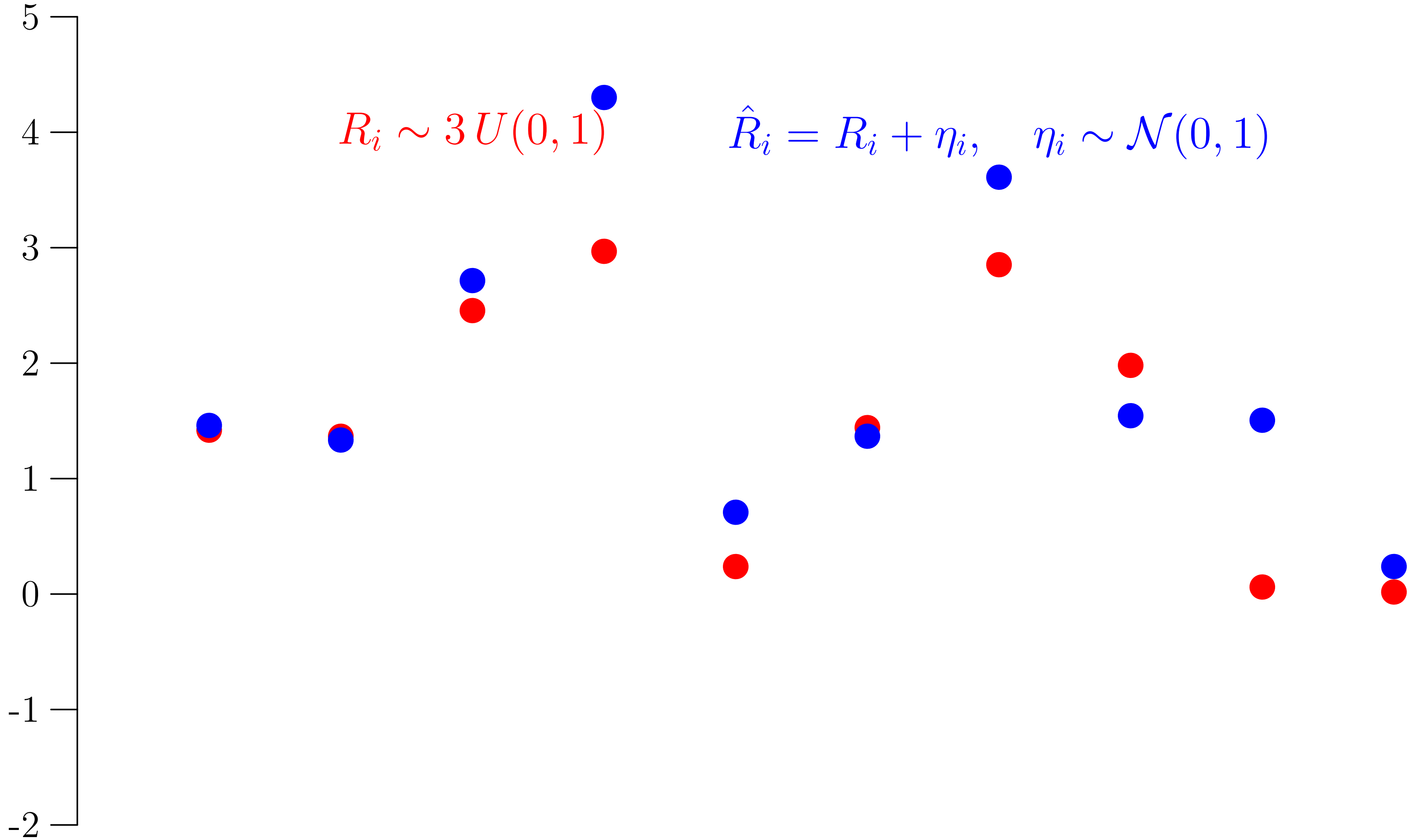
Sins of MNIST

- MNIST isn't that hard (MLPs can do a reasonable job)
- The digits are already centred and rescaled so it is not that interesting
- Many practitioner's would run test multiple times and state their best performance
- For most good networks the task is so easy that the errors are where the scaling and centring went wrong or the digits are unintelligible
- Ignoring this practitioners would start over-fitting on the test set

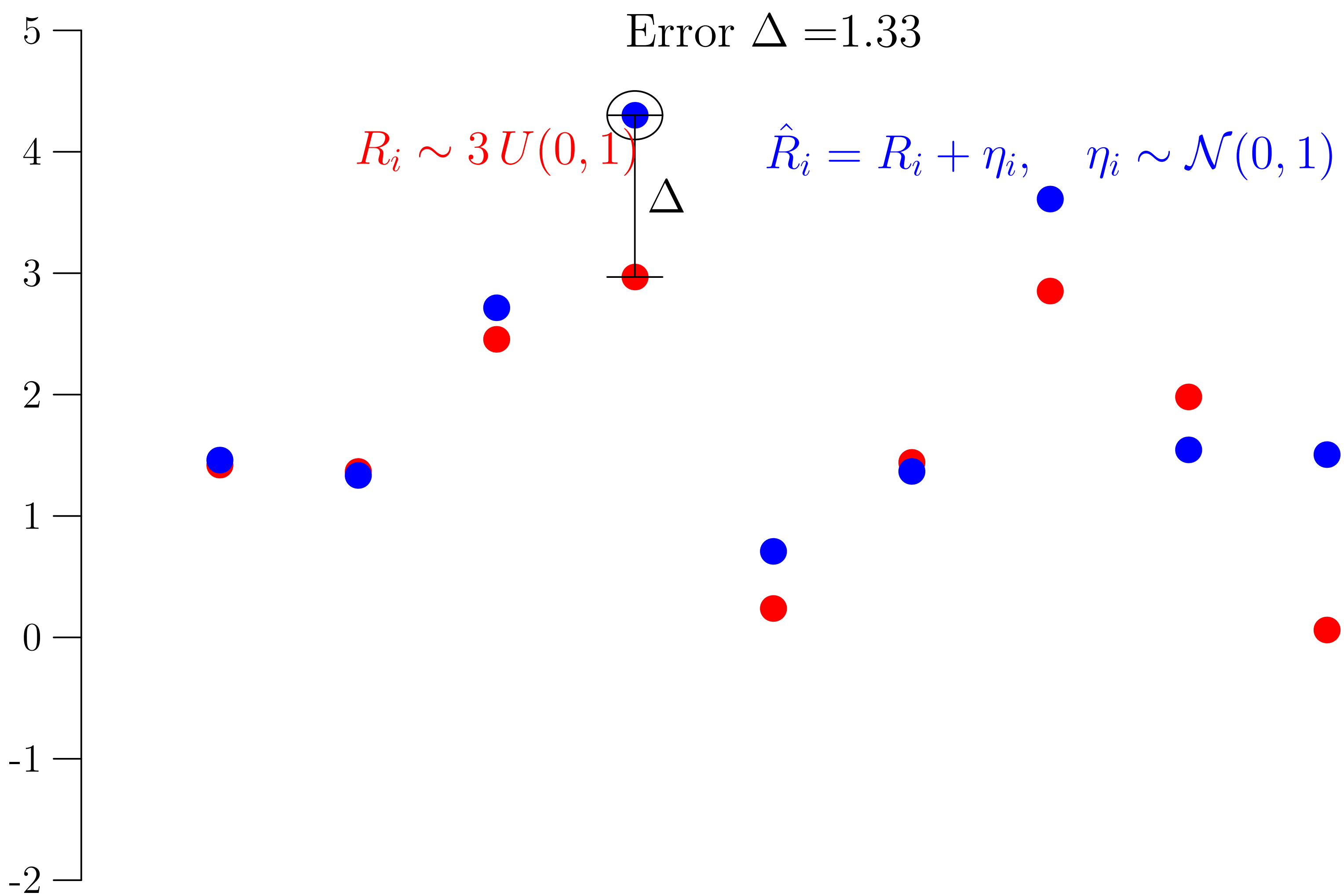
The Over-fitting Game



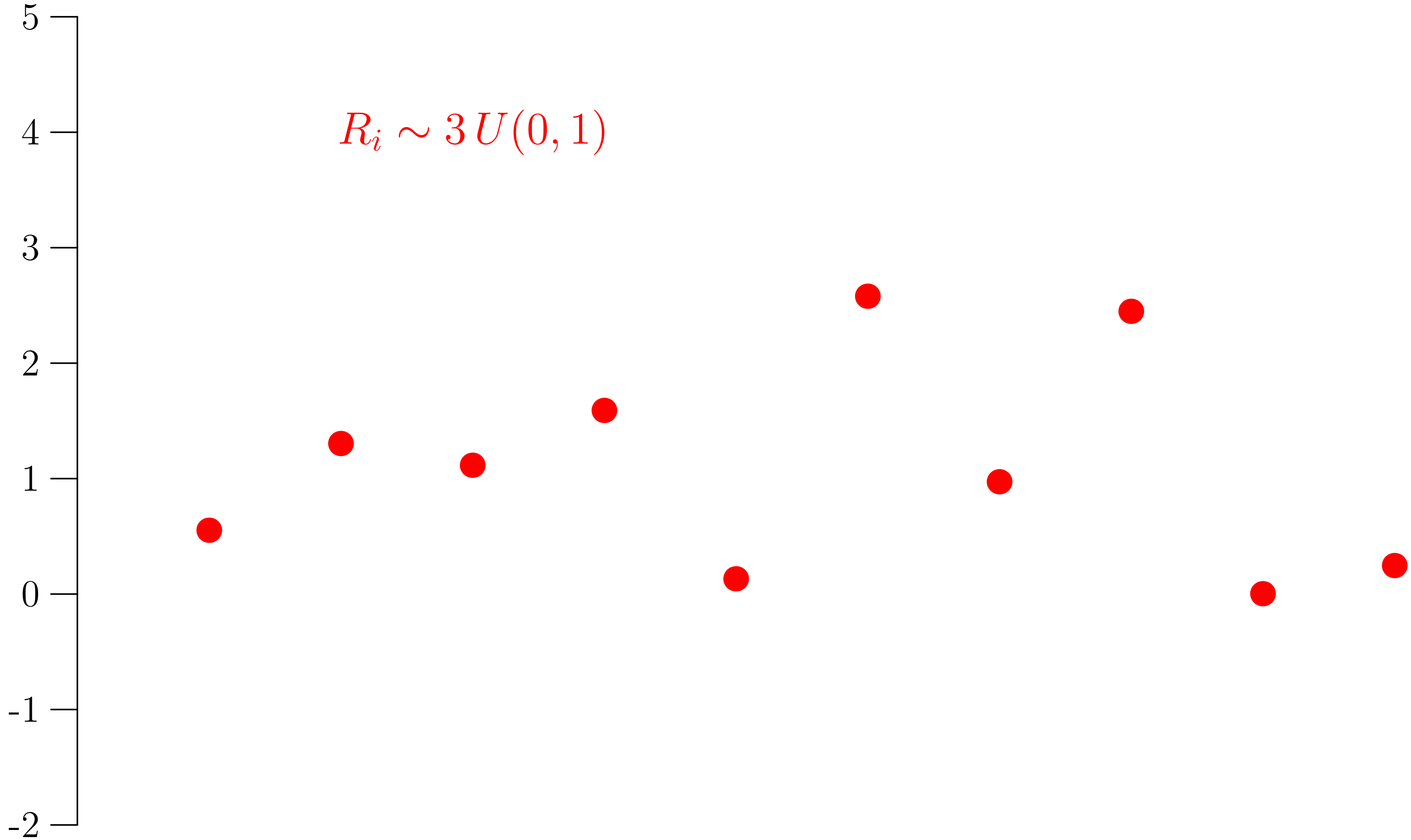
The Over-fitting Game



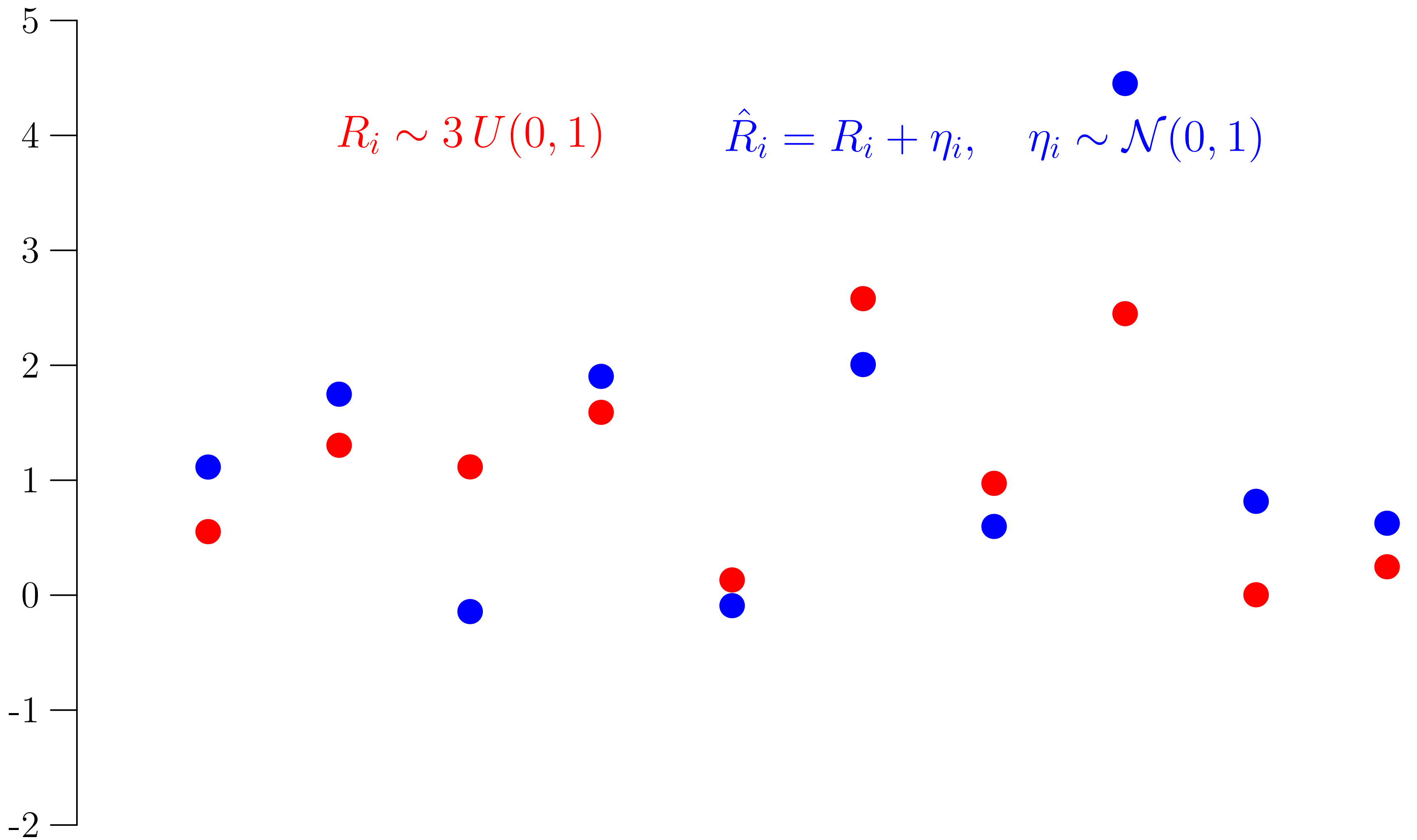
The Over-fitting Game



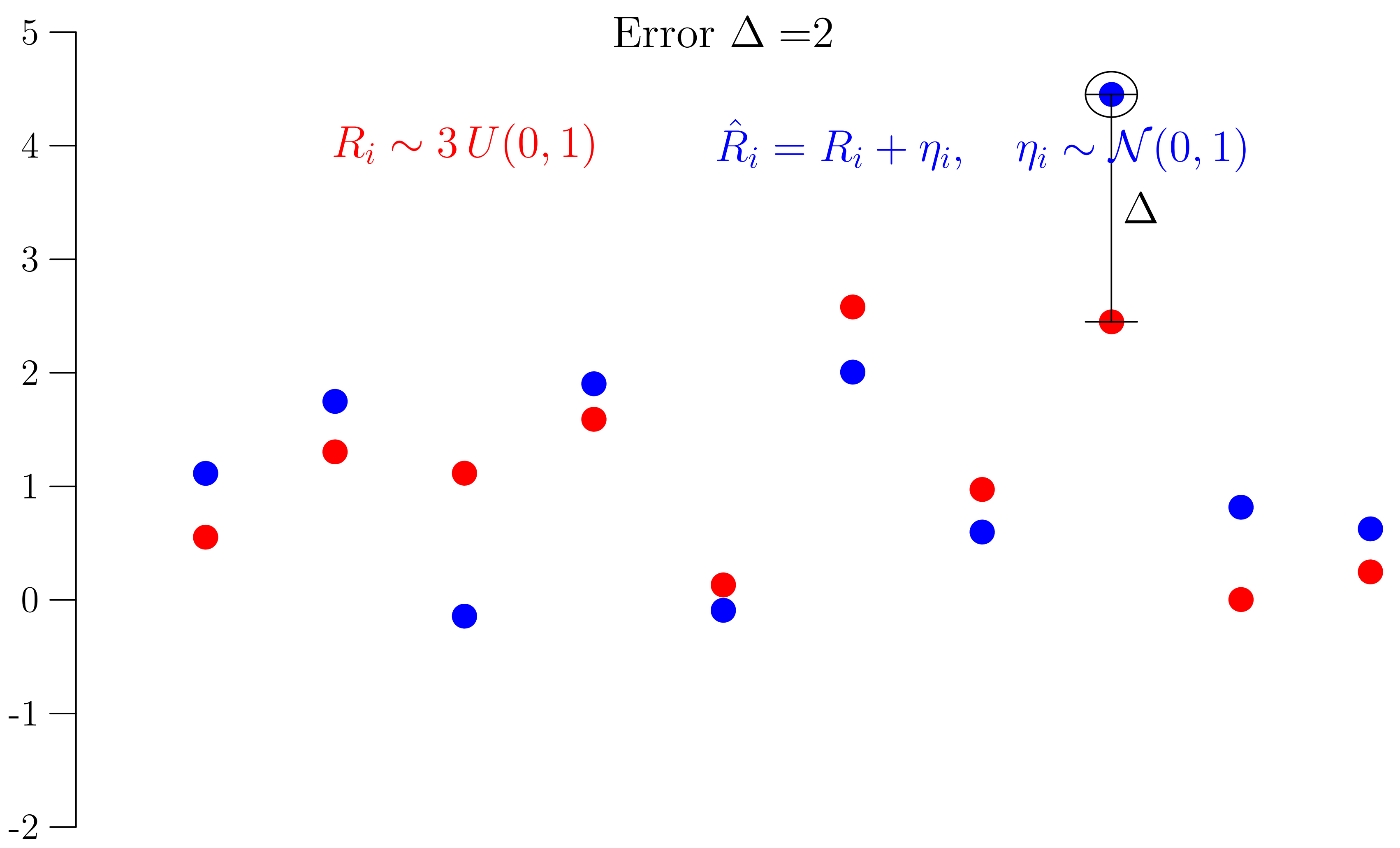
The Over-fitting Game



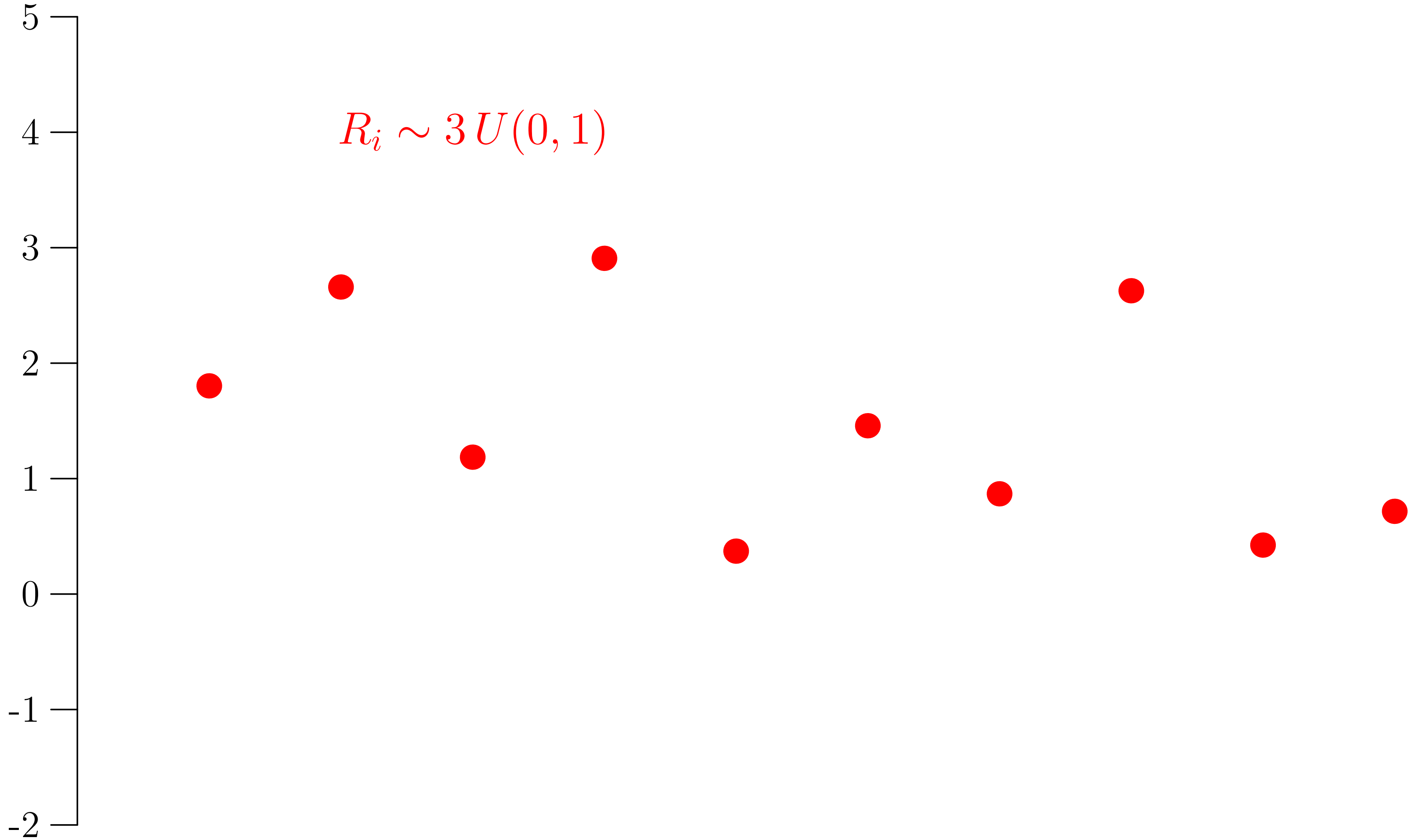
The Over-fitting Game



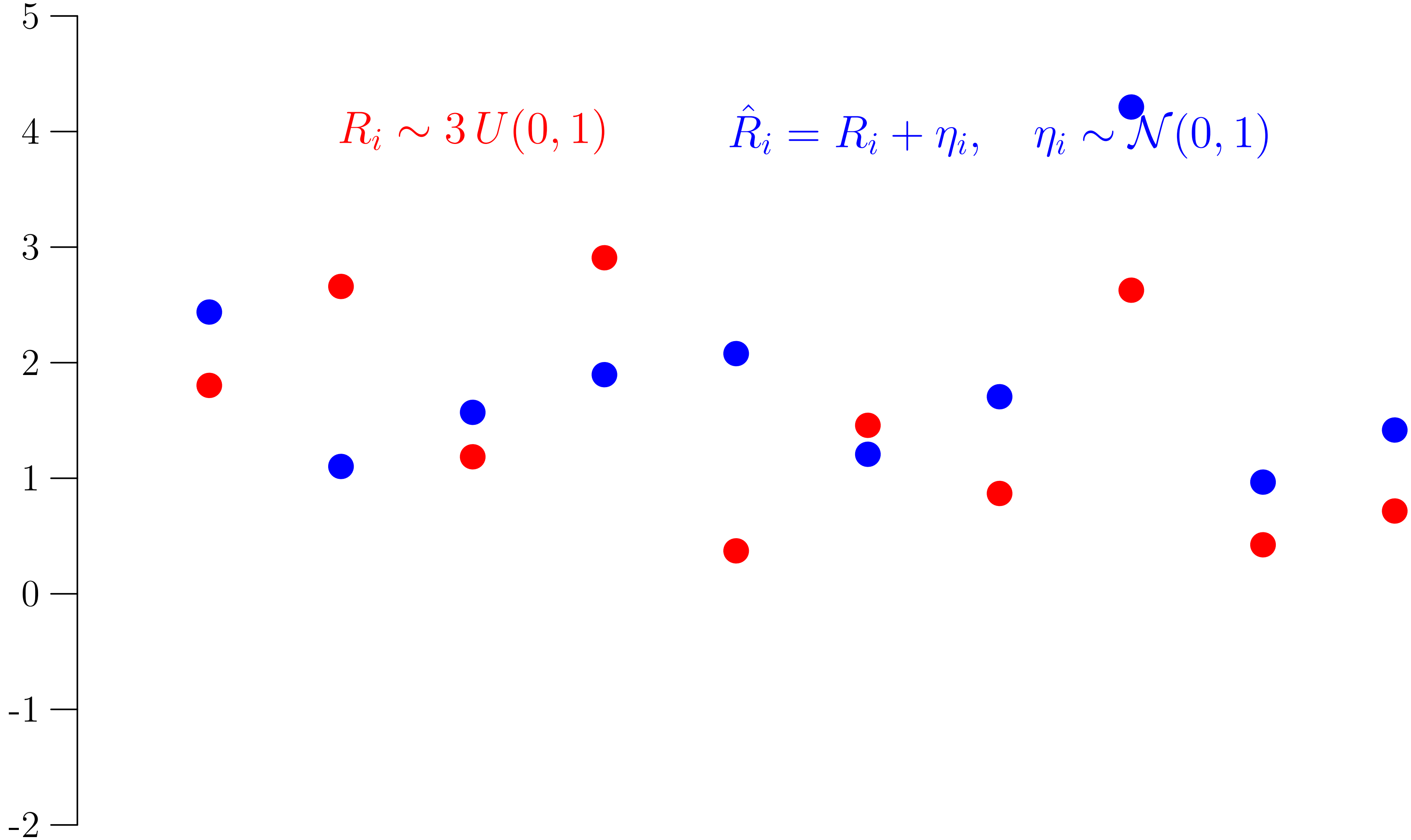
The Over-fitting Game



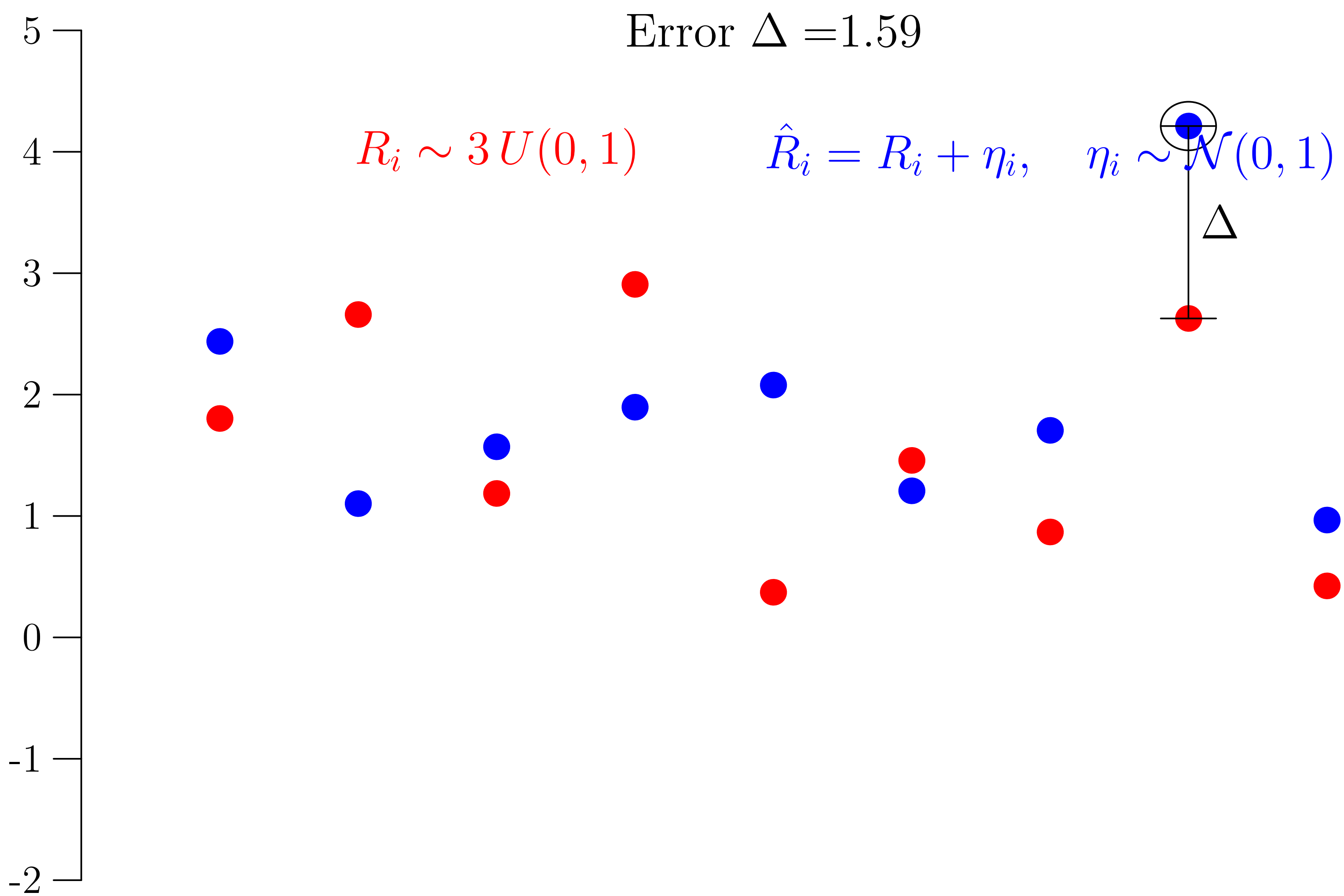
The Over-fitting Game



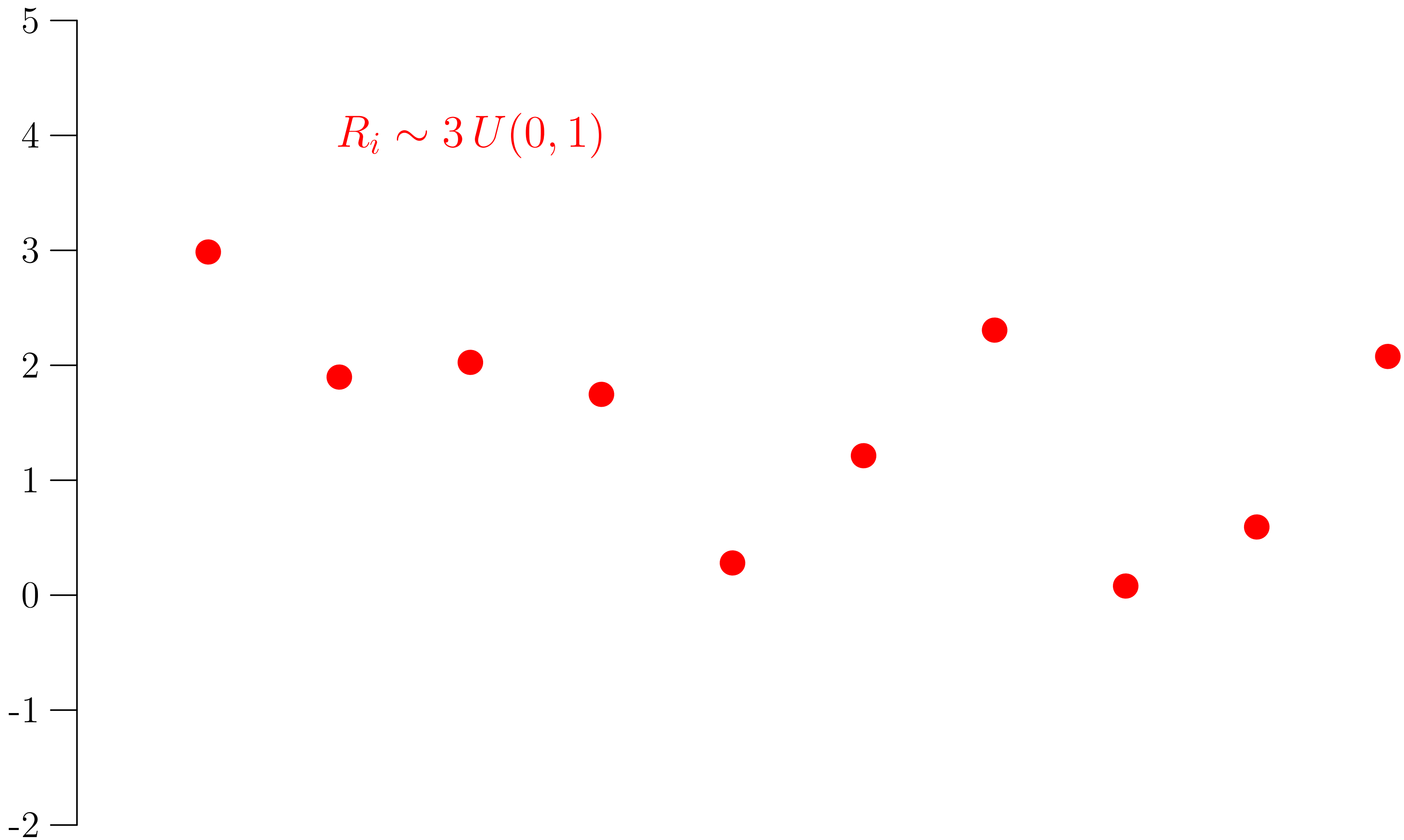
The Over-fitting Game



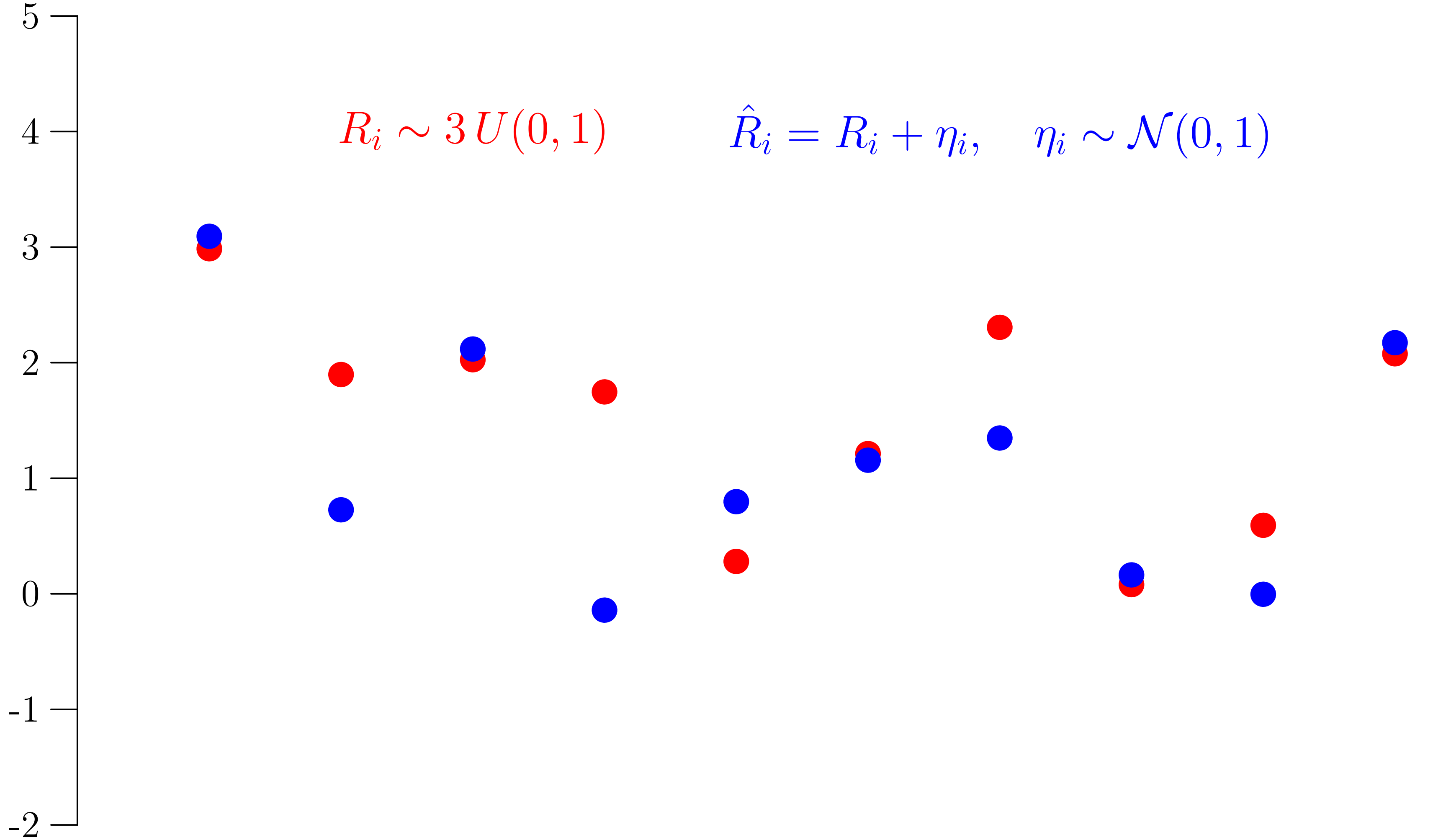
The Over-fitting Game



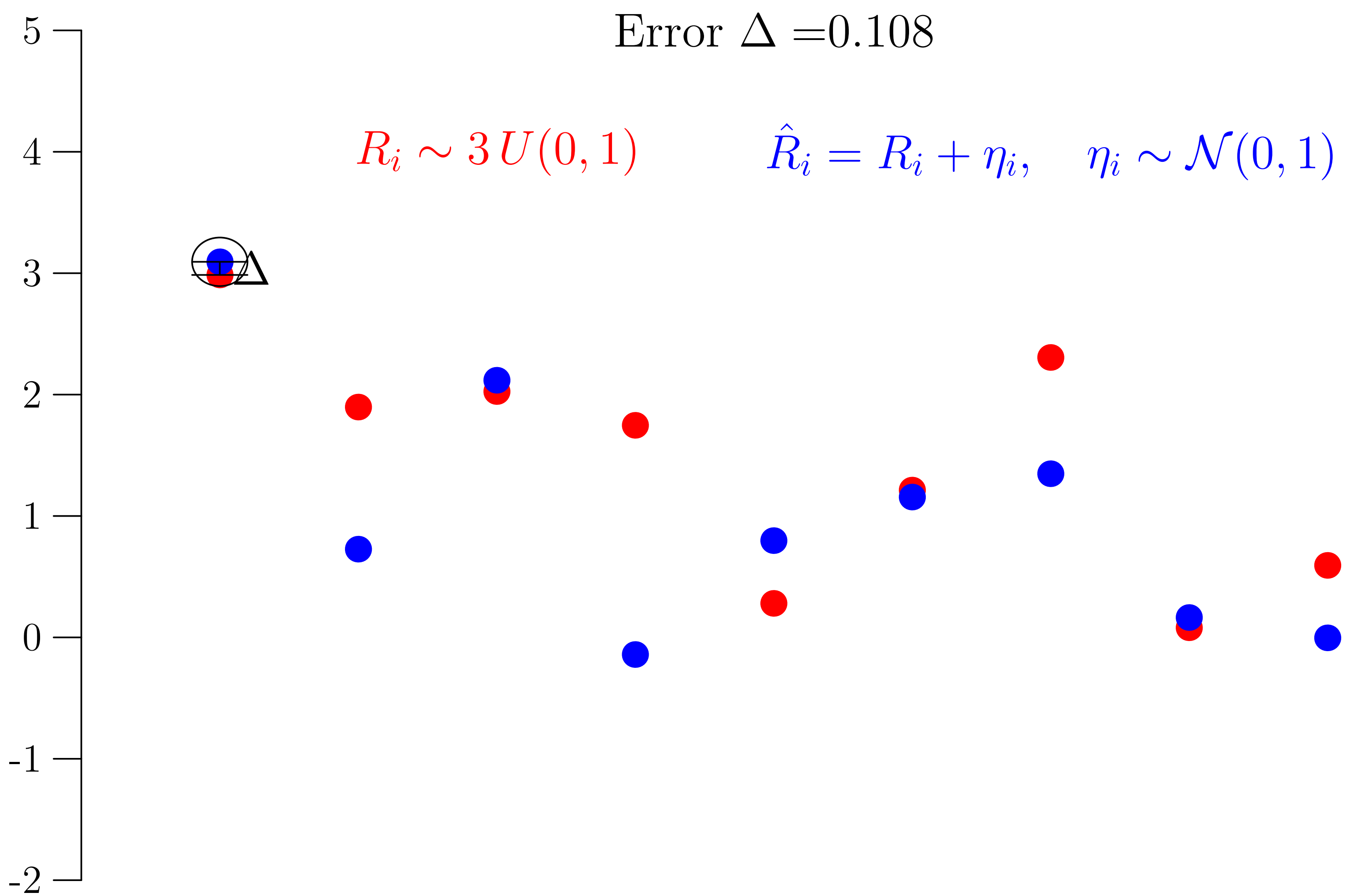
The Over-fitting Game



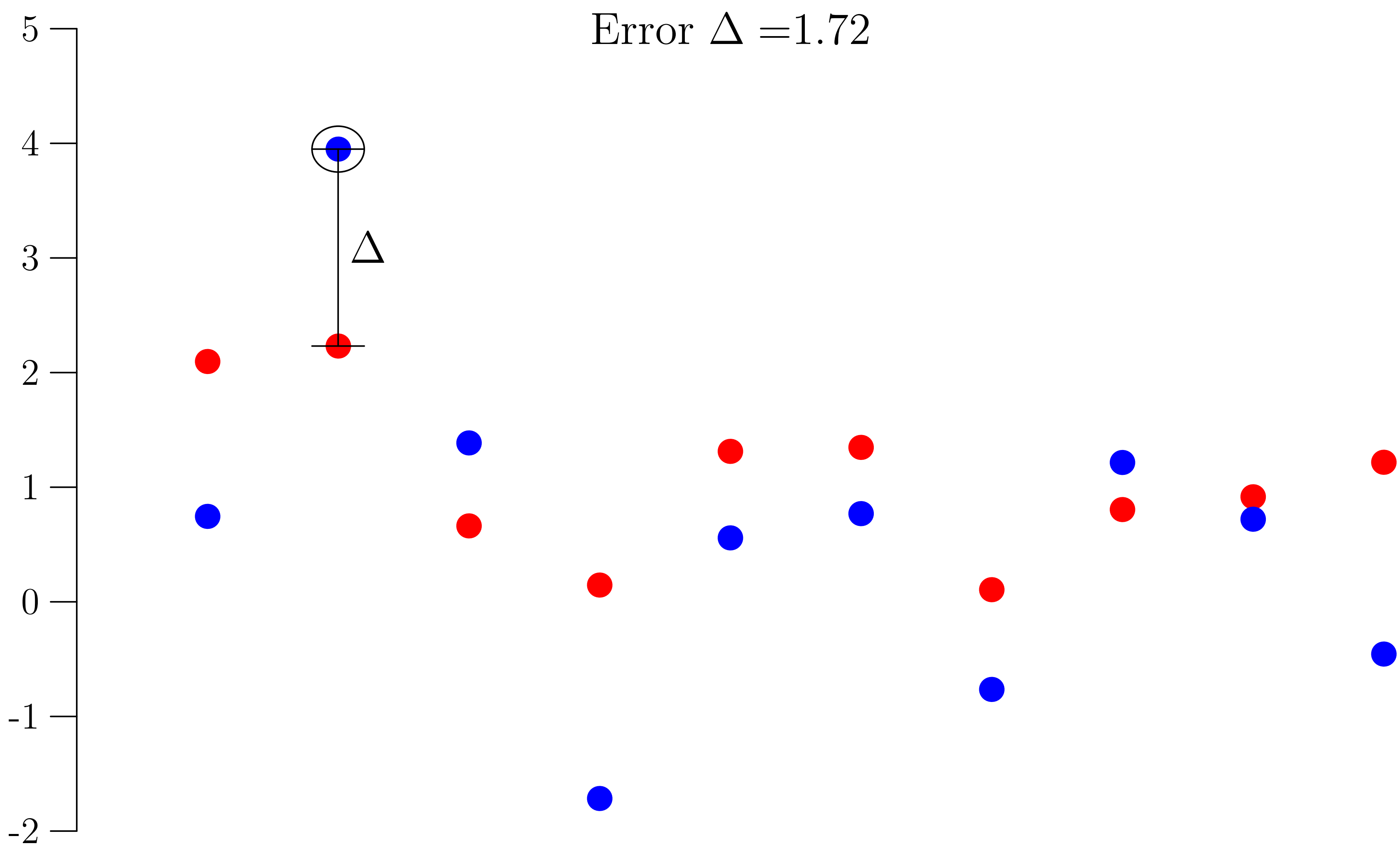
The Over-fitting Game



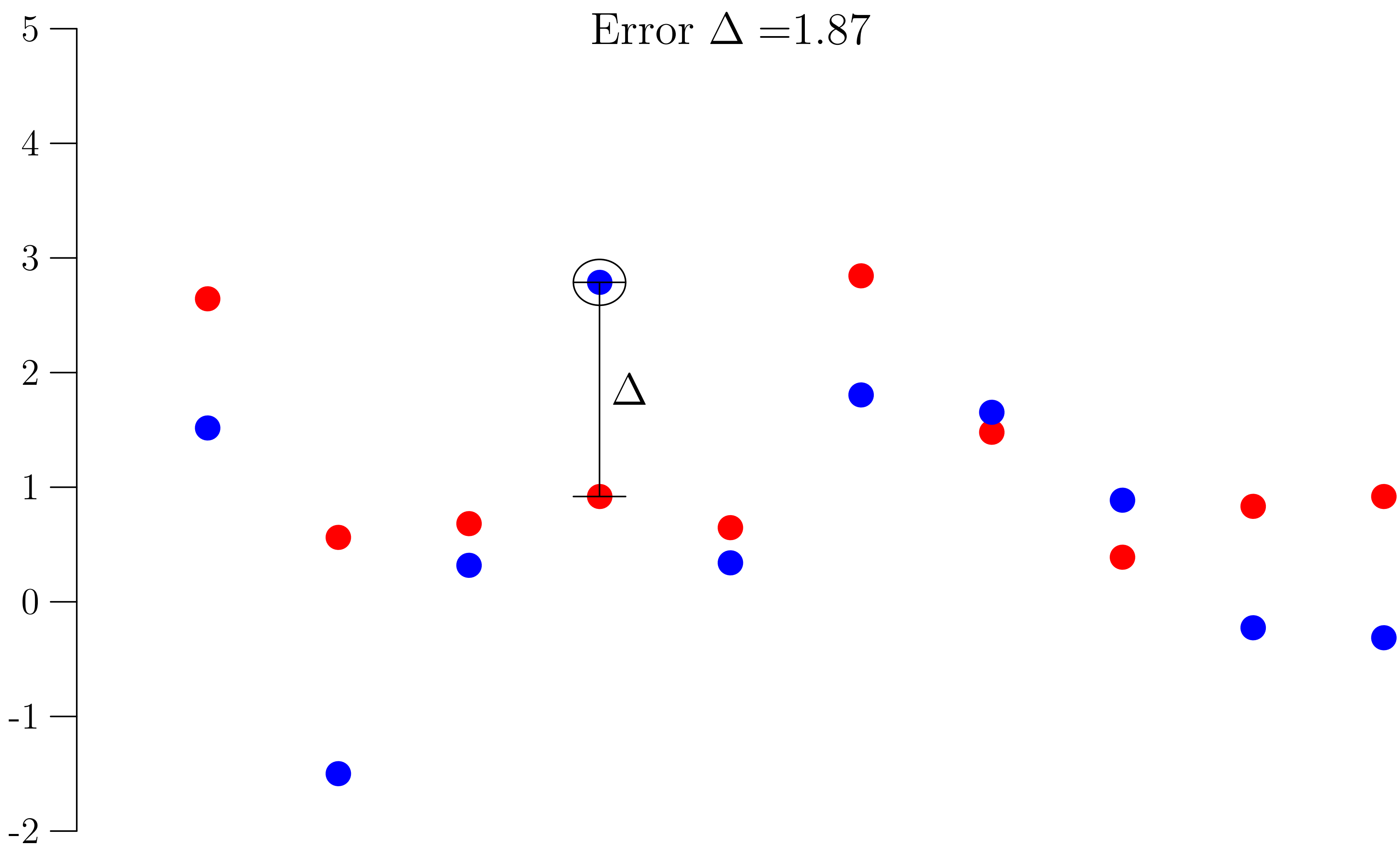
The Over-fitting Game



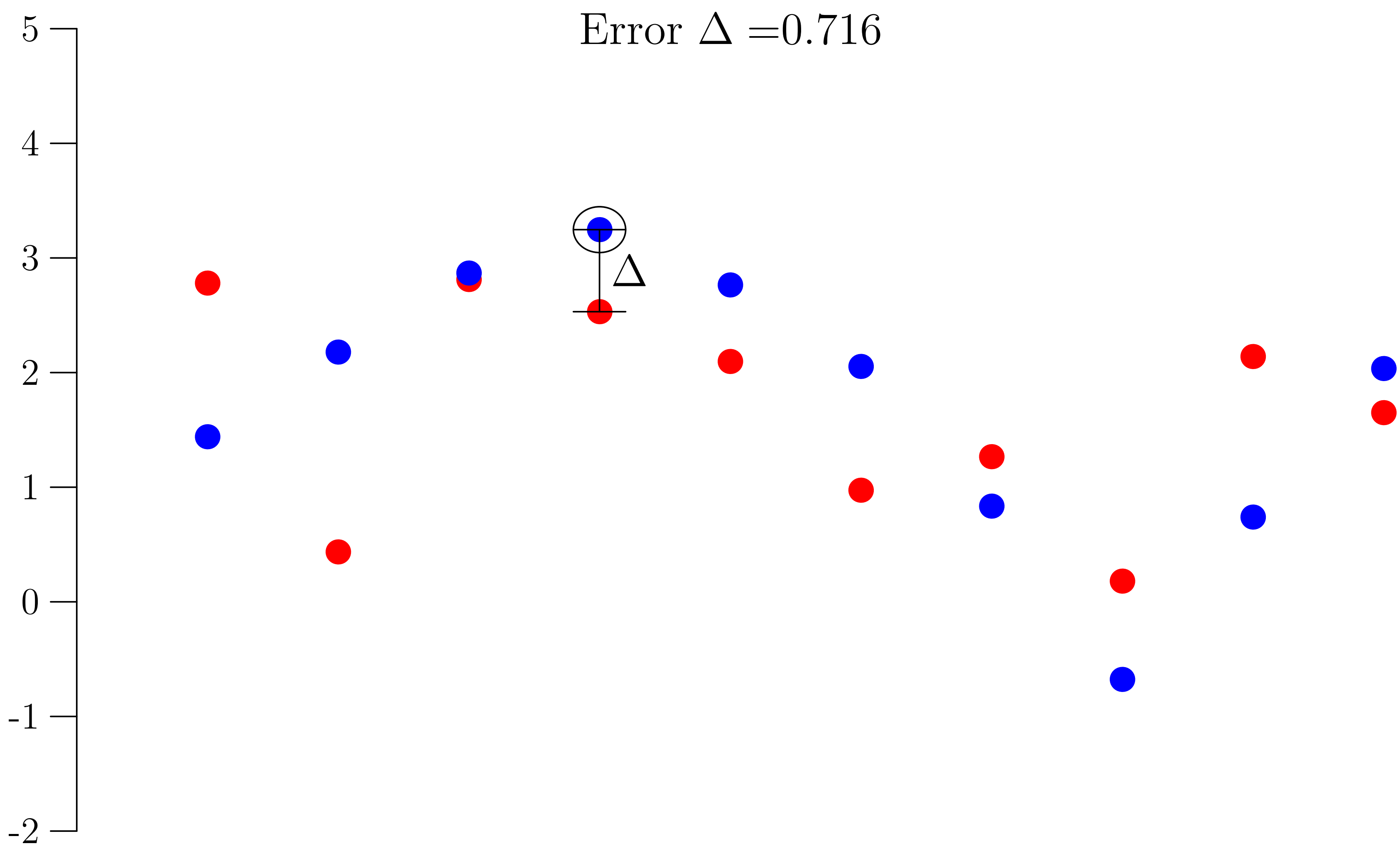
The Over-fitting Game



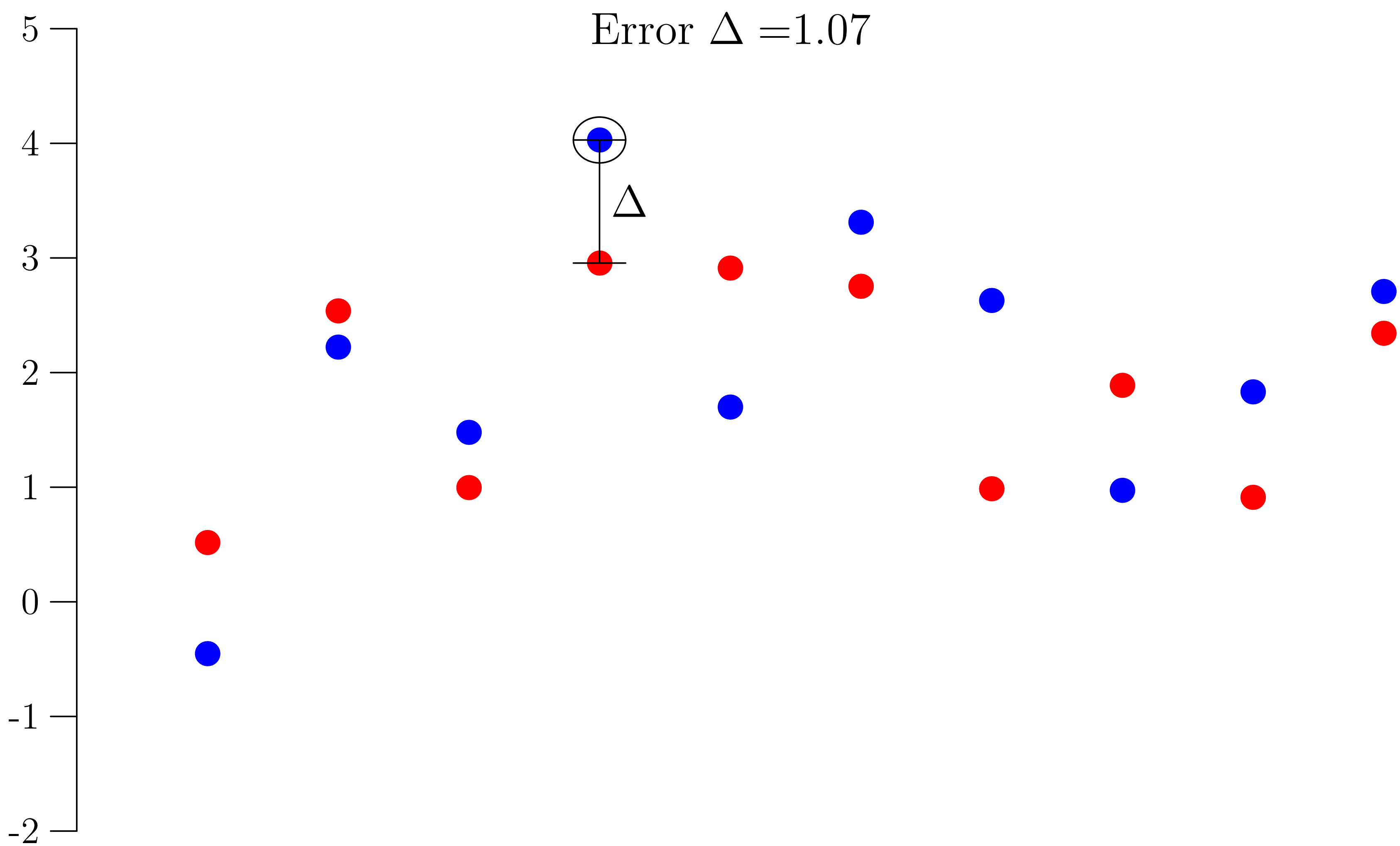
The Over-fitting Game



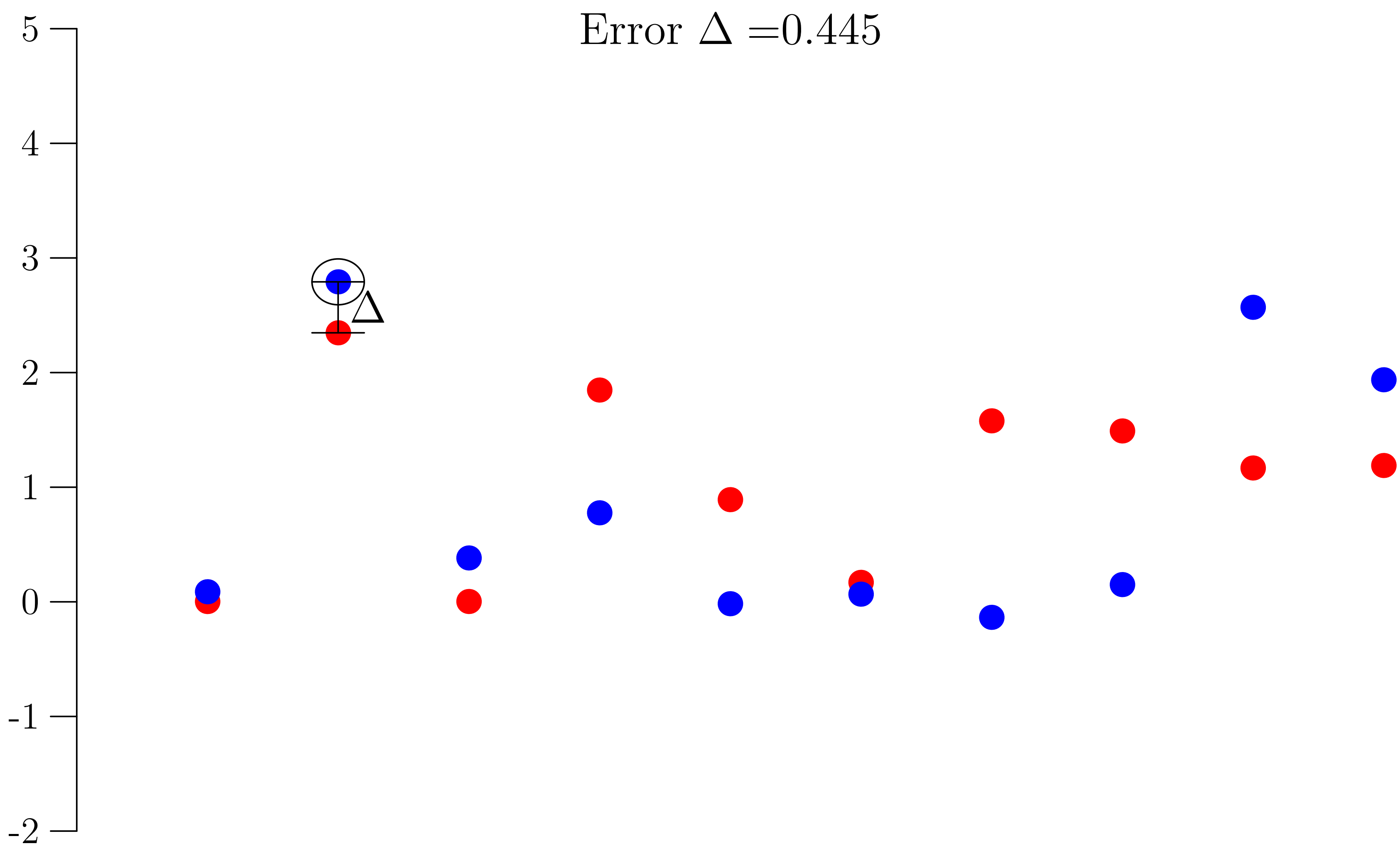
The Over-fitting Game



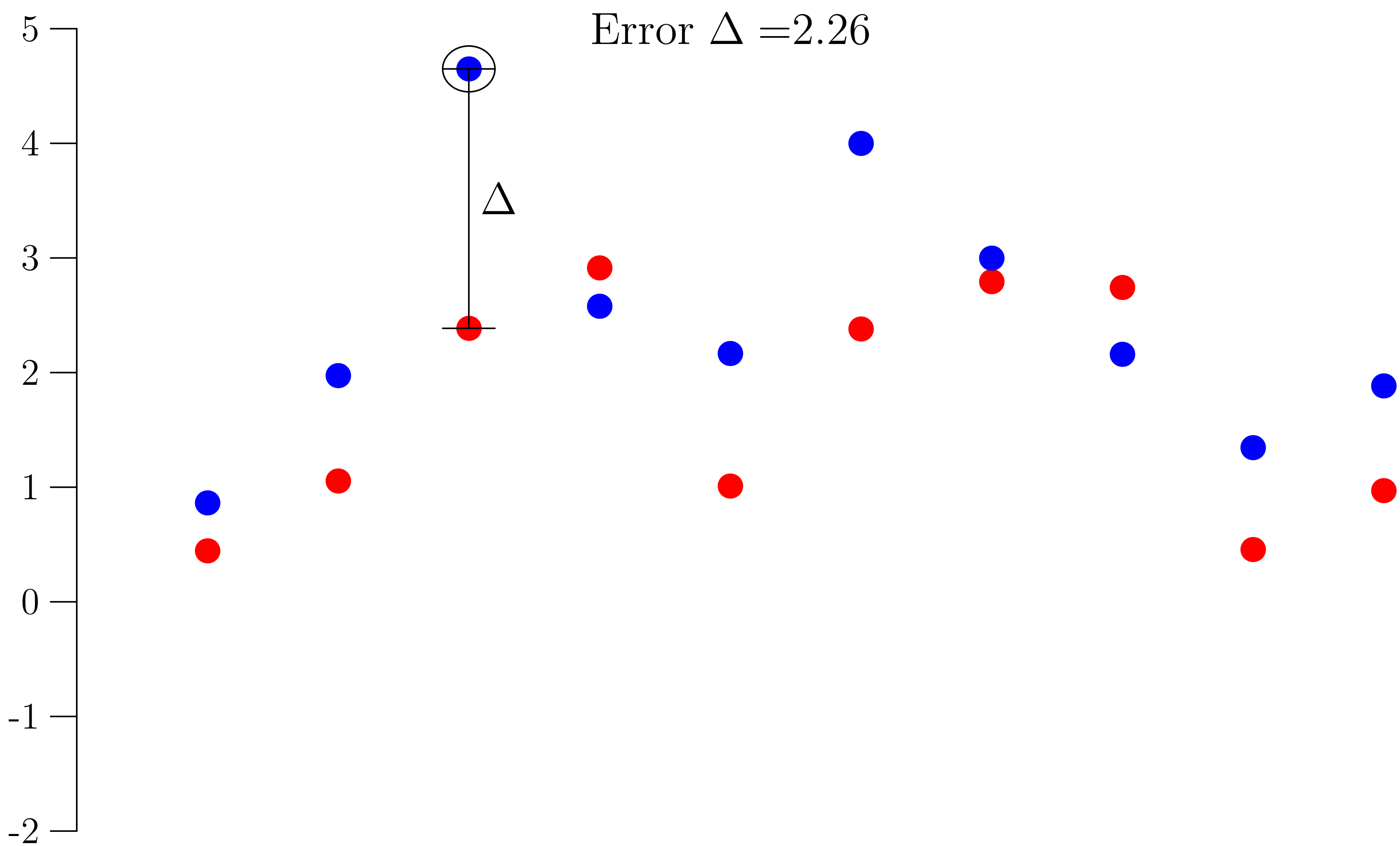
The Over-fitting Game



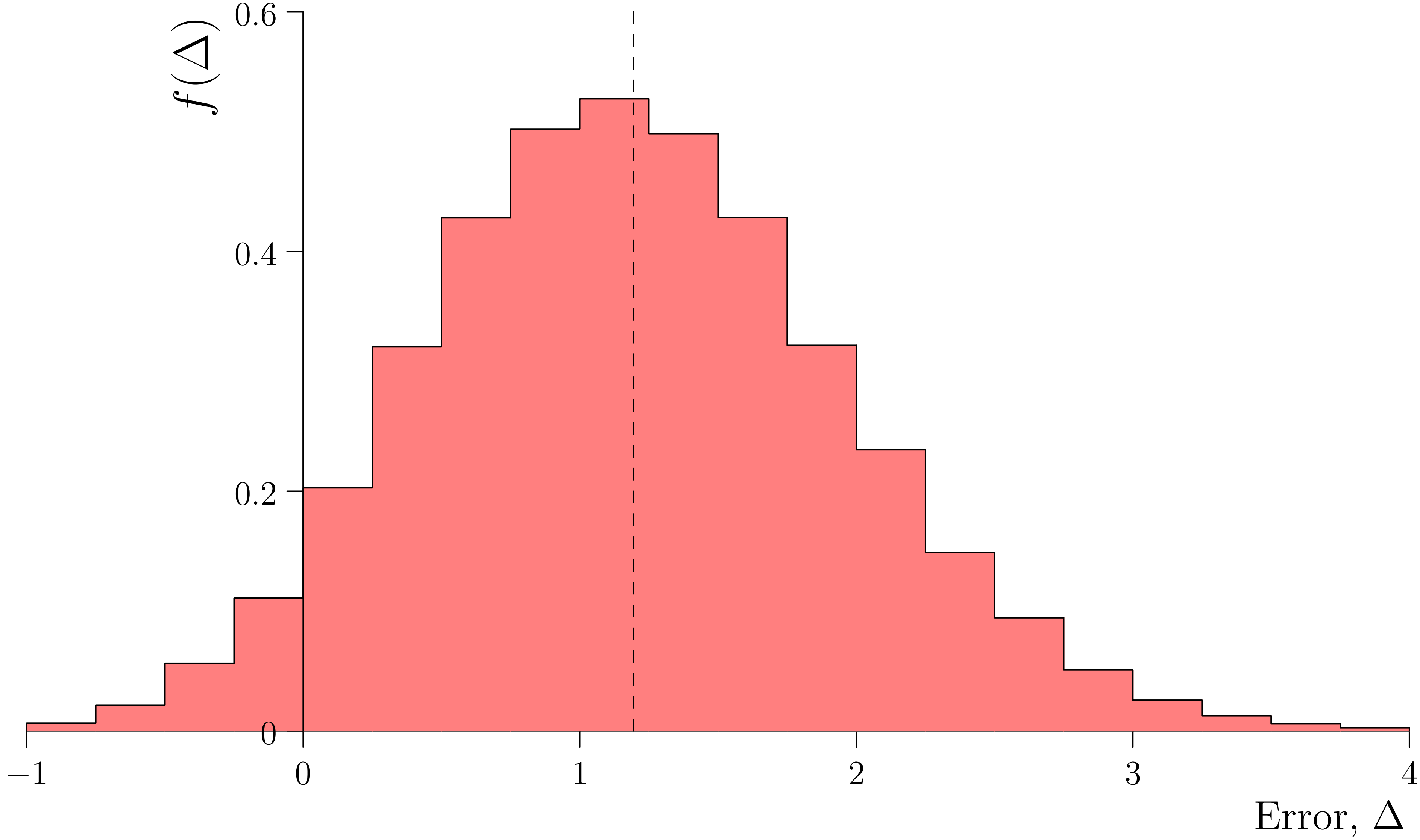
The Over-fitting Game



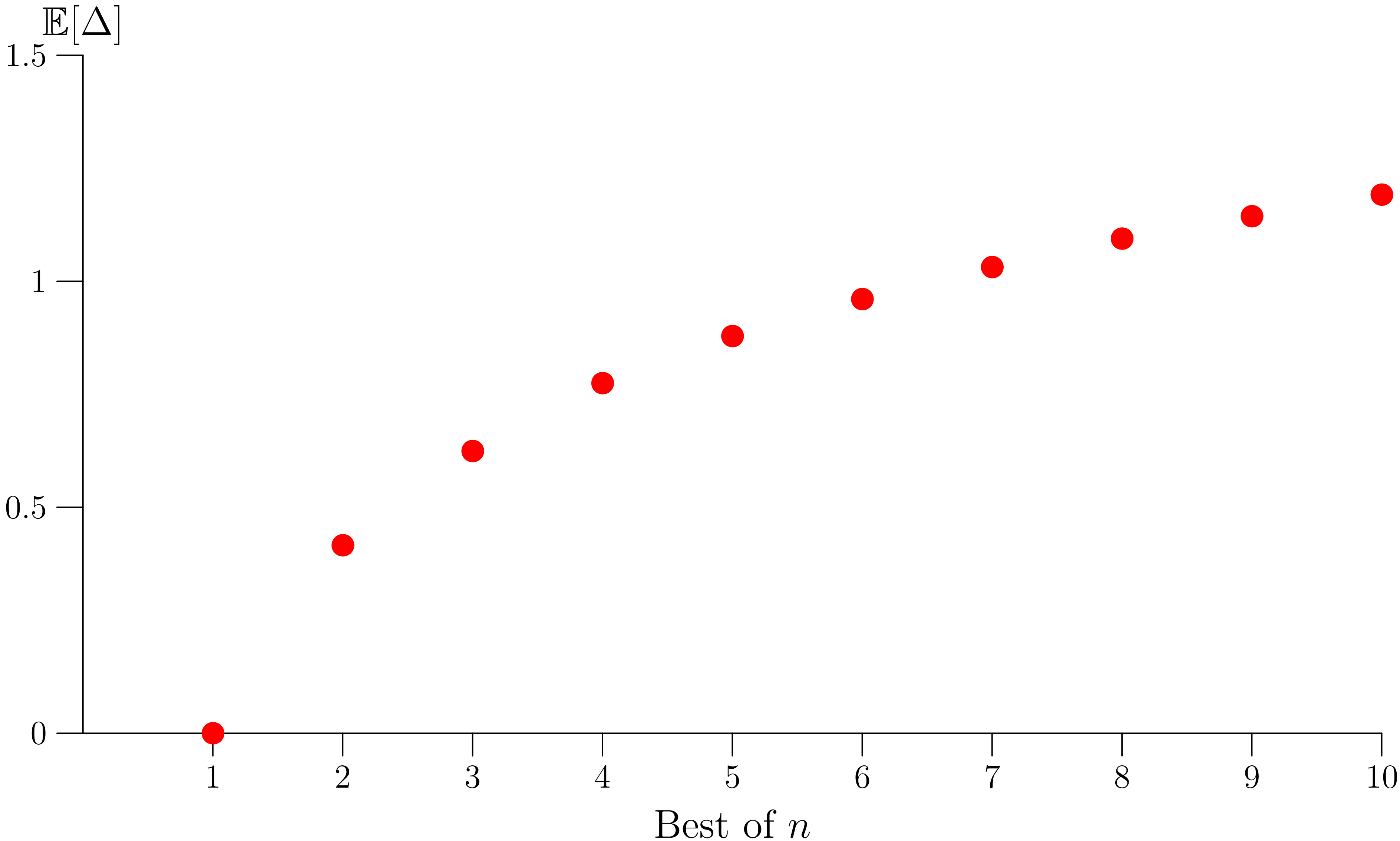
The Over-fitting Game



The Over-fitting Game



The Over-fitting Game



Overfitting the Test Set

- If you try to improve your performance by regularly testing on your test set you quickly overfit to the test set
- In many Kaggle competitions there is a test set provided early on which is used to maintain a leader board
- At the end of the competition, the competitors are tested on a new test set and the order of the leader board is radically changed
- You can be honest by using a separate **validation set**

Overfitting the Test Set

- If you try to improve your performance by regularly testing on your test set you quickly overfit to the test set

- In many Kaggle competitions there is a test set provided early on which is used to maintain a leader board

- At the end of the competition, the competitors are tested on a new test set and the order of the leader board is radically changed

- You can be honest by using a separate **validation set**

Overfitting the Test Set

- If you try to improve your performance by regularly testing on your test set you quickly overfit to the test set
- In many Kaggle competitions there is a test set provided early on which is used to maintain a leader board
- At the end of the competition, the competitors are tested on a new test set and the order of the leader board is radically changed
- You can be honest by using a separate **validation set**

Overfitting the Test Set

- If you try to improve your performance by regularly testing on your test set you quickly overfit to the test set

- In many Kaggle competitions there is a test set provided early on which is used to maintain a leader board

- At the end of the competition, the competitors are tested on a new test set and the order of the leader board is radically changed

- You can be honest by using a separate **validation set**

Overfitting the Test Set

- If you try to improve your performance by regularly testing on your test set you quickly overfit to the test set
- In many Kaggle competitions there is a test set provided early on which is used to maintain a leader board
- At the end of the competition, the competitors are tested on a new test set and the order of the leader board is radically changed
- You can be honest by using a separate **validation set**
- But that doesn't stop you from over-fitting the validation set

Overfitting the Test Set

- If you try to improve your performance by regularly testing on your test set you quickly overfit to the test set
- In many Kaggle competitions there is a test set provided early on which is used to maintain a leader board
- At the end of the competition, the competitors are tested on a new test set and the order of the leader board is radically changed
- You can be honest by using a separate **validation set**
- But that doesn't stop you from over-fitting the validation set
- **Lesson 1: Beware of over-fitting on the test set**

CIFAR-10

- Moving beyond MNIST, practitioners turned to CIFAR-10
- 10 class problem with colour images of size 32×32
- MLPs struggle

CIFAR-10

- Moving beyond MNIST, practitioners turned to CIFAR-10
- 10 class problem with colour images of size 32×32
- MLPs struggle

CIFAR-10

- Moving beyond MNIST, practitioners turned to CIFAR-10
- 10 class problem with colour images of size 32×32
- MLPs struggle

Dodgy Test Sets

- 10 or so years after CIFAR-10 was produce Benjamin Recht *et al.* constructed a new test set for CIFAR-10 using the same protocol
- The performance of the best performing models dropped from 4% error to 10% error
- A lot of this was down to over-fitting the test set
- But it also turn out there was some contamination in the original test-set with images in the training set appearing in the test set
- There may have been some *shortcut learning* (perhaps trucks were taken predominantly from a web site with images from mobile phones, birds being taken with long lens)

Dodgy Test Sets

- 10 or so years after CIFAR-10 was produce Benjamin Recht *et al.* constructed a new test set for CIFAR-10 using the same protocol
- The performance of the best performing models dropped from 4% error to 10% error
- A lot of this was down to over-fitting the test set
- But it also turn out there was some contamination in the original test-set with images in the training set appearing in the test set
- There may have been some *shortcut learning* (perhaps trucks were taken predominantly from a web site with images from mobile phones, birds being taken with long lens)

Dodgy Test Sets

- 10 or so years after CIFAR-10 was produce Benjamin Recht *et al.* constructed a new test set for CIFAR-10 using the same protocol
- The performance of the best performing models dropped from 4% error to 10% error
- A lot of this was down to over-fitting the test set
- But it also turn out there was some contamination in the original test-set with images in the training set appearing in the test set
- There may have been some *shortcut learning* (perhaps trucks were taken predominantly from a web site with images from mobile phones, birds being taken with long lens)

Dodgy Test Sets

- 10 or so years after CIFAR-10 was produce Benjamin Recht *et al.* constructed a new test set for CIFAR-10 using the same protocol
- The performance of the best performing models dropped from 4% error to 10% error
- A lot of this was down to over-fitting the test set
- But it also turn out there was some contamination in the original test-set with images in the training set appearing in the test set
- There may have been some *shortcut learning* (perhaps trucks were taken predominantly from a web site with images from mobile phones, birds being taken with long lens)

Dodgy Test Sets

- 10 or so years after CIFAR-10 was produce Benjamin Recht *et al.* constructed a new test set for CIFAR-10 using the same protocol
- The performance of the best performing models dropped from 4% error to 10% error
- A lot of this was down to over-fitting the test set
- But it also turn out there was some contamination in the original test-set with images in the training set appearing in the test set
- There may have been some *shortcut learning* (perhaps trucks were taken predominantly from a web site with images from mobile phones, birds being taken with long lens)

Dodgy Test Sets

- 10 or so years after CIFAR-10 was produce Benjamin Recht *et al.* constructed a new test set for CIFAR-10 using the same protocol
- The performance of the best performing models dropped from 4% error to 10% error
- A lot of this was down to over-fitting the test set
- But it also turn out there was some contamination in the original test-set with images in the training set appearing in the test set
- There may have been some *shortcut learning* (perhaps trucks were taken predominantly from a web site with images from mobile phones, birds being taken with long lens)
- **Lesson 2: Be suspicious of data sets**

ImageNet

- ImageNet for several years was considered the Gold standard for measuring image classification performance

- The ImageNet competition started in 2009 was instrumental in starting the deep learning revolution with AlexNet beating the competition in 2011 by 10% and ushered in GPUs

- It used large images (224×224 or 256×256) with 1000 classes and 1.3 million training examples

ImageNet

- ImageNet for several years was considered the Gold standard for measuring image classification performance

- The ImageNet competition started in 2009 was instrumental in starting the deep learning revolution with AlexNet beating the competition in 2011 by 10% and ushered in GPUs

- It used large images (224×224 or 256×256) with 1000 classes and 1.3 million training examples

ImageNet

- ImageNet for several years was considered the Gold standard for measuring image classification performance
- The ImageNet competition started in 2009 was instrumental in starting the deep learning revolution with AlexNet beating the competition in 2011 by 10% and ushered in GPUs
- It used large images (224×224 or 256×256) with 1000 classes and 1.3 million training examples

ImageNet

- They stopped the classification competition in 2017 to stop people overfitting on the testset
- The dataset requires a lot of compute to train a network: practioners often use mini-ImageNet versions to reduce heating up the planet
- It has oddity, humans exist in the image, but are ignored (there is no human category)
- There is no perfect training set, but ImageNet pushed the field more than most

ImageNet

- They stopped the classification competition in 2017 to stop people overfitting on the testset
- The dataset requires a lot of compute to train a network: practioners often use mini-ImageNet versions to reduce heating up the planet
- It has oddity, humans exist in the image, but are ignored (there is no human category)
- There is no perfect training set, but ImageNet pushed the field more than most

ImageNet

- They stopped the classification competition in 2017 to stop people overfitting on the testset
- The dataset requires a lot of compute to train a network: practioners often use mini-ImageNet versions to reduce heating up the planet
- It has oddity, humans exist in the image, but are ignored (there is no human category)
- There is no perfect training set, but ImageNet pushed the field more than most

ImageNet

- They stopped the classification competition in 2017 to stop people overfitting on the testset
- The dataset requires a lot of compute to train a network: practioners often use mini-ImageNet versions to reduce heating up the planet
- It has oddity, humans exist in the image, but are ignored (there is no human category)
- There is no perfect training set, but ImageNet pushed the field more than most

ImageNet

- They stopped the classification competition in 2017 to stop people overfitting on the testset
- The dataset requires a lot of compute to train a network: practioners often use mini-ImageNet versions to reduce heating up the planet
- It has oddity, humans exist in the image, but are ignored (there is no human category)
- There is no perfect training set, but ImageNet pushed the field more than most

Data Set Contamination

- Because training large models is time consuming it is regular practice to build on fine-tuned large (foundation) models
- This is especially true in LLMs
- Sometimes it is possible to get **surprisingly** good performance on tasks because the backbone model was trained on that task
- If you want to measure true performance you need to be strict in ensuring that the test set is truely independent of the training set

Data Set Contamination

- Because training large models is time consuming it is regular practice to build on fine-tuned large (foundation) models
- This is especially true in LLMs
- Sometimes it is possible to get **surprisingly** good performance on tasks because the backbone model was trained on that task
- If you want to measure true performance you need to be strict in ensuring that the test set is truely independent of the training set

Data Set Contamination

- Because training large models is time consuming it is regular practice to build on fine-tuned large (foundation) models
- This is especially true in LLMs
- Sometimes it is possible to get **surprisingly** good performance on tasks because the backbone model was trained on that task
- If you want to measure true performance you need to be strict in ensuring that the test set is truely independent of the training set

Data Set Contamination

- Because training large models is time consuming it is regular practice to build on fine-tuned large (foundation) models
- This is especially true in LLMs
- Sometimes it is possible to get **surprisingly** good performance on tasks because the backbone model was trained on that task
- If you want to measure true performance you need to be strict in ensuring that the test set is truely independent of the training set

Data Set Contamination

- Because training large models is time consuming it is regular practice to build on fine-tuned large (foundation) models
- This is especially true in LLMs
- Sometimes it is possible to get **surprisingly** good performance on tasks because the backbone model was trained on that task
- If you want to measure true performance you need to be strict in ensuring that the test set is truely independent of the training set
- **Lesson 3: Avoid data set contamination**

Short Cut Learning

- The field is full of datasets where there is a short-cut for making a classification
- Classic examples include X-ray diagnostic systems that pick-up on post-operative artifacts showing the patient had a disease
- Machine learning will pick-up on any consistent signal that separates data classes, even when these are not what you want to learn

Short Cut Learning

- The field is full of datasets where there is a short-cut for making a classification
- Classic examples include X-ray diagnostic systems that pick-up on post-operative artifacts showing the patient had a disease
- Machine learning will pick-up on any consistent signal that separates data classes, even when these are not what you want to learn

Short Cut Learning

- The field is full of datasets where there is a short-cut for making a classification
- Classic examples include X-ray diagnostic systems that pick-up on post-operative artifacts showing the patient had a disease
- Machine learning will pick-up on any consistent signal that separates data classes, even when these are not what you want to learn

Short Cut Learning

- The field is full of datasets where there is a short-cut for making a classification

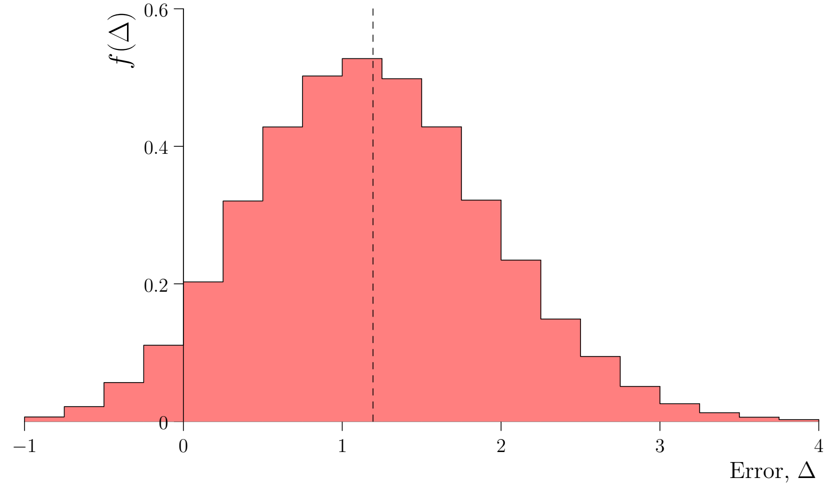
- Classic examples include X-ray diagnostic systems that pick-up on post-operative artifacts showing the patient had a disease

- Machine learning will pick-up on any consistent signal that separates data classes, even when these are not what you want to learn

- **Lesson 4: When results seem too good to be true, often it is too good to be true**

Outline

1. Data Sets
2. **Reproducibility**
3. Metrics



Look at My New Model

- Machine Learning has always be plagued by people developing new models
- Sometimes these model turn out to change everything (AlexNet, Transformers)
- More often these models are claimed to beat an existing model, but vanish without a bubble
- There are many causes

Look at My New Model

- Machine Learning has always be plagued by people developing new models
- Sometimes these model turn out to change everything (AlexNet, Transformers)
- More often these models are claimed to beat an existing model, but vanish without a bubble
- There are many causes

Look at My New Model

- Machine Learning has always be plagued by people developing new models
- Sometimes these model turn out to change everything (AlexNet, Transformers)
- More often these models are claimed to beat an existing model, but vanish without a bubble
- There are many causes

Look at My New Model

- Machine Learning has always be plagued by people developing new models
- Sometimes these model turn out to change everything (AlexNet, Transformers)
- More often these models are claimed to beat an existing model, but vanish without a bubble
- There are many causes

One-trick Pony

- Models are often tested on a single problem
- Either they are compared against a weak baseline
- Or they have been optimised, while the baseline used out-of-the-box hyper-parameters
- Referees have become very wary of these claims and will often insist on testing on multiple datasets, against strong baselines that are equally optimised
- This is boring to do, but comes from experience

One-trick Pony

- Models are often tested on a single problem
- Either they are compared against a weak baseline
- Or they have been optimised, while the baseline used out-of-the-box hyper-parameters
- Referees have become very wary of these claims and will often insist on testing on multiple datasets, against strong baselines that are equally optimised
- This is boring to do, but comes from experience

One-trick Pony

- Models are often tested on a single problem
- Either they are compared against a weak baseline
- Or they have been optimised, while the baseline used out-of-the-box hyper-parameters
- Referees have become very wary of these claims and will often insist on testing on multiple datasets, against strong baselines that are equally optimised
- This is boring to do, but comes from experience

One-trick Pony

- Models are often tested on a single problem
- Either they are compared against a weak baseline
- Or they have been optimised, while the baseline used out-of-the-box hyper-parameters
- Referees have become very wary of these claims and will often insist on testing on multiple datasets, against strong baselines that are equally optimised
- This is boring to do, but comes from experience

One-trick Pony

- Models are often tested on a single problem
- Either they are compared against a weak baseline
- Or they have been optimised, while the baseline used out-of-the-box hyper-parameters
- Referees have become very wary of these claims and will often insist on testing on multiple datasets, against strong baselines that are equally optimised
- This is boring to do, but comes from experience

One-trick Pony

- Models are often tested on a single problem
- Either they are compared against a weak baseline
- Or they have been optimised, while the baseline used out-of-the-box hyper-parameters
- Referees have become very wary of these claims and will often insist on testing on multiple datasets, against strong baselines that are equally optimised
- This is boring to do, but comes from experience
- **Lesson 5: Be rigorous in comparing with others**

Irreproducibility

- Many results claimed in papers cannot be reproduced
- A classic example happened to a much vaunted paper by Geoff Hinton about an architecture know as Capsules claiming a SOTA performance of 0.25% test error on MNIST
- This was never reproduced
- ML went through an existential crisis about reproducing results
- It is now standard practice to publish code to allow others to quickly verify your results

Irreproducibility

- Many results claimed in papers cannot be reproduced
- A classic example happened to a much vaunted paper by Geoff Hinton about an architecture know as Capsules claiming a SOTA performance of 0.25% test error on MNIST
- This was never reproduced
- ML went through an existential crisis about reproducing results
- It is now standard practice to publish code to allow others to quickly verify your results

Irreproducibility

- Many results claimed in papers cannot be reproduced
- A classic example happened to a much vaunted paper by Geoff Hinton about an architecture know as Capsules claiming a SOTA performance of 0.25% test error on MNIST
- This was never reproduced
- ML went through an existential crisis about reproducing results
- It is now standard practice to publish code to allow others to quickly verify your results

Irreproducibility

- Many results claimed in papers cannot be reproduced
- A classic example happened to a much vaunted paper by Geoff Hinton about an architecture know as Capsules claiming a SOTA performance of 0.25% test error on MNIST
- This was never reproduced
- ML went through an existential crisis about reproducing results
- It is now standard practice to publish code to allow others to quickly verify your results

Irreproducibility

- Many results claimed in papers cannot be reproduced
- A classic example happened to a much vaunted paper by Geoff Hinton about an architecture know as Capsules claiming a SOTA performance of 0.25% test error on MNIST
- This was never reproduced
- ML went through an existential crisis about reproducing results
- It is now standard practice to publish code to allow others to quickly verify your results

Irreproducibility

- Many results claimed in papers cannot be reproduced
- A classic example happened to a much vaunted paper by Geoff Hinton about an architecture know as Capsules claiming a SOTA performance of 0.25% test error on MNIST
- This was never reproduced
- ML went through an existential crisis about reproducing results
- It is now standard practice to publish code to allow others to quickly verify your results
- **Lesson 6: Make your work reproducible: prepare to make your code public**

Claims on How it Works

- Machine Learning papers often make convincing claims about why models work
- To avoid this and convince colleagues it is now standard practice to deconstruct your model by removing the parts that are believe crucial and measuring the change in performance
- These are known as *ablation* studies and are now standard in most papers

Claims on How it Works

- Machine Learning papers often make convincing claims about why models work, which turn out to be wrong
- To avoid this and convince colleagues it is now standard practice to deconstruct your model by removing the parts that are believe crucial and measuring the change in performance
- These are known as $ablation$ studies and are now standard in most papers

Claims on How it Works

- Machine Learning papers often make convincing claims about why models work, which turn out to be wrong
- To avoid this and convince colleagues it is now standard practice to deconstruct your model by removing the parts that are believe crucial and measuring the change in performance
- These are known as $ablation$ studies and are now standard in most papers

Claims on How it Works

- Machine Learning papers often make convincing claims about why models work, which turn out to be wrong
- To avoid this and convince colleagues it is now standard practice to deconstruct your model by removing the parts that are believe crucial and measuring the change in performance
- These are known as $ablation$ studies and are now standard in most papers

Claims on How it Works

- Machine Learning papers often make convincing claims about why models work, which turn out to be wrong
- To avoid this and convince colleagues it is now standard practice to deconstruct your model by removing the parts that are believe crucial and measuring the change in performance
- These are known as $ablation$ studies and are now standard in most papers
- **Lesson 7: Perform ablation studies where meaningful**

Theories of How ML Works

- More generally plausible theories of how ML/DL work rapidly build a following
- These theories are backed up by experiments according to a particular protocol
- These theory are often only true in a very limited domain

Theories of How ML Works

- More generally plausible theories of how ML/DL work rapidly build a following
- These theories are backed up by experiments according to a particular protocol
- These theory are often only true in a very limited domain

Theories of How ML Works

- More generally plausible theories of how ML/DL work rapidly build a following
- These theories are backed up by experiments according to a particular protocol
- These theory are often only true in a very limited domain

Theories of How ML Works

- More generally plausible theories of how ML/DL work rapidly build a following
- These theories are backed up by experiments according to a particular protocol
- These theory are often only true in a very limited domain, or over-simplification

Theories of How ML Works

- More generally plausible theories of how ML/DL work rapidly build a following
- These theories are backed up by experiments according to a particular protocol
- These theory are often only true in a very limited domain, or over-simplification, or just plain wrong

Theories of How ML Works

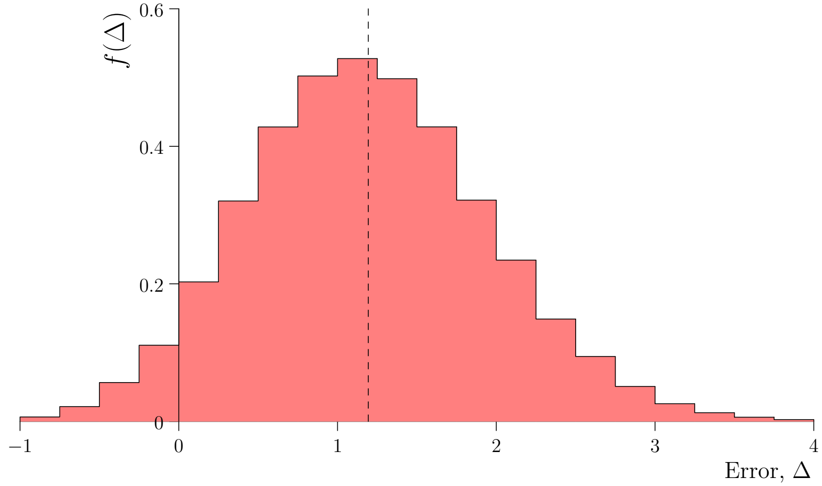
- More generally plausible theories of how ML/DL work rapidly build a following
- These theories are backed up by experiments according to a particular protocol
- These theory are often only true in a very limited domain, or over-simplification, or just plain wrong
- **Lesson 8: Beware of jumping on band-waggons and rigorously test theories empirically**

Theories of How ML Works

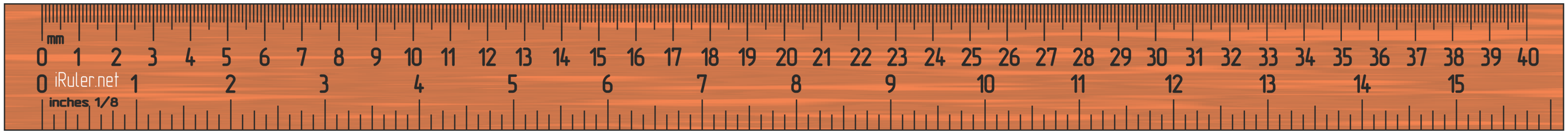
- More generally plausible theories of how ML/DL work rapidly build a following
- These theories are backed up by experiments according to a particular protocol
- These theory are often only true in a very limited domain, or over-simplification, or just plain wrong
- **Lesson 8: Beware of jumping on band-waggons and rigorously test theories empirically even when there is a large community accepting the theory**

Outline

1. Data Sets
2. Reproducibility
3. **Metrics**

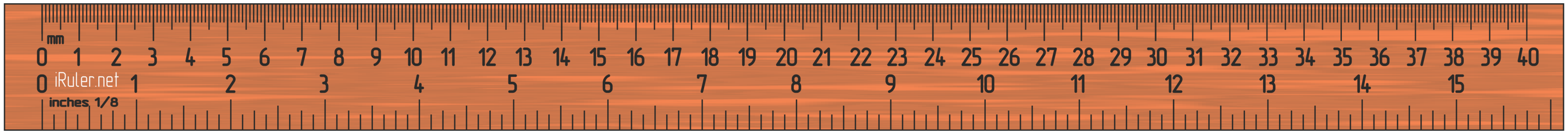


Metrics



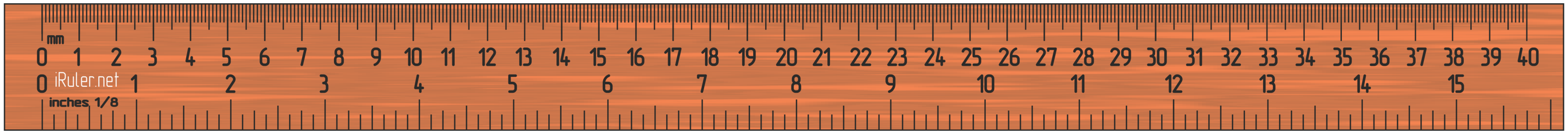
- One of the most important and often most challenging jobs in machine learning is to measure performance of a learning machine
- For classic classification and regression tasks this is usually given
- For a lot of modern, semi-supervised tasks it can be a research project in itself to find a suitable metric

Metrics



- One of the most important and often most challenging jobs in machine learning is to measure performance of a learning machine
- For classic classification and regression tasks this is usually given
- For a lot of modern, semi-supervised tasks it can be a research project in itself to find a suitable metric

Metrics



- One of the most important and often most challenging jobs in machine learning is to measure performance of a learning machine
- For classic classification and regression tasks this is usually given
- For a lot of modern, semi-supervised tasks it can be a research project in itself to find a suitable metric

Choosing Good Metrics

- The is a plethora of metrics that exist (e.g. F1-score, BLEU, Rouge, FID, . . .)
- If you are building on previous work, you may need to stick with published metrics
- For many tasks one metric is usually insufficient to capture all relevant aspects of a problem

Choosing Good Metrics

- The is a plethora of metrics that exist (e.g. F1-score, BLEU, Rouge, FID, . . .)
- If you are building on previous work, you may need to stick with published metrics
- For many tasks one metric is usually insufficient to capture all relevant aspects of a problem

Choosing Good Metrics

- The is a plethora of metrics that exist (e.g. F1-score, BLEU, Rouge, FID, . . .)
- If you are building on previous work, you may need to stick with published metrics
- For many tasks one metric is usually insufficient to capture all relevant aspects of a problem

Choosing Good Metrics

- The is a plethora of metrics that exist (e.g. F1-score, BLEU, Rouge, FID, . . .)
- If you are building on previous work, you may need to stick with published metrics
- For many tasks one metric is usually insufficient to capture all relevant aspects of a problem
- **Lesson 9: Think carefully about what you want to measure and use appropriate metrics**

Choosing Good Metrics

- The is a plethora of metrics that exist (e.g. F1-score, BLEU, Rouge, FID, . . .)
- If you are building on previous work, you may need to stick with published metrics
- For many tasks one metric is usually insufficient to capture all relevant aspects of a problem
- **Lesson 9: Think carefully about what you want to measure and use appropriate metrics**
- **Lesson 10: Help readers to understand you metrics (should it be maximised or minimised? What values are acceptable/good/bad?)**

Human Metrics

- For some problems a suitable metric is human evaluation
- When this does not involve specialists then there are many crowd-sourcing companies that can do this
- This is expensive, so practitioners often fall-back of automatic metrics
- These surrogates can be rather poor

Human Metrics

- For some problems a suitable metric is human evaluation
- When this does not involve specialists then there are many crowd-sourcing companies that can do this
- This is expensive, so practitioners often fall-back of automatic metrics
- These surrogates can be rather poor

Human Metrics

- For some problems a suitable metric is human evaluation
- When this does not involve specialists then there are many crowd-sourcing companies that can do this
- This is expensive, so practitioners often fall-back of automatic metrics
- These surrogates can be rather poor

Human Metrics

- For some problems a suitable metric is human evaluation
- When this does not involve specialists then there are many crowd-sourcing companies that can do this
- This is expensive, so practitioners often fall-back of automatic metrics
- These surrogates can be rather poor

Human Metrics

- For some problems a suitable metric is human evaluation
- When this does not involve specialists then there are many crowd-sourcing companies that can do this
- This is expensive, so practitioners often fall-back of automatic metrics
- These surrogates can be rather poor—they might be the best you can do with a computer

Human Metrics

- For some problems a suitable metric is human evaluation
- When this does not involve specialists then there are many crowd-sourcing companies that can do this
- This is expensive, so practitioners often fall-back of automatic metrics
- These surrogates can be rather poor—they might be the best you can do with a computer, but be aware of their weaknesses

Don't Necessarily Follow the Trend

- A lot of machine learning research focuses on improving some metric
- Often the performance on this metric is actually good enough and the real problem lies elsewhere
- Thinking about what people really want often leads to far more productive research

Don’t Necessarily Follow the Trend

- A lot of machine learning research focuses on improving some metric
- Often the performance on this metric is actually good enough and the real problem lies elsewhere
- Thinking about what people really want often leads to far more productive research

Don't Necessarily Follow the Trend

- A lot of machine learning research focuses on improving some metric
- Often the performance on this metric is actually good enough and the real problem lies elsewhere
- Thinking about what people really want often leads to far more productive research

Don't Necessarily Follow the Trend

- A lot of machine learning research focuses on improving some metric
- Often the performance on this metric is actually good enough and the real problem lies elsewhere
- Thinking about what people really want often leads to far more productive research
- **Lesson 11: When everyone is concentrated on improving one metric, think is there another metric that measures something that may be more important**

Presenting Metrics

- Measuring performance comes with errors
- There may be systematic error due to the biases in your test set
- There will also be statistical errors coming from using a finite random test set
- Interpreting empirical metrics without errors is dangerous

Presenting Metrics

- Measuring performance comes with errors
- There may be systematic error due to the biases in your test set
- There will also be statistical errors coming from using a finite random test set
- Interpreting empirical metrics without errors is dangerous

Presenting Metrics

- Measuring performance comes with errors
- There may be systematic error due to the biases in your test set—think carefully about is your test fair
- There will also be statistical errors coming from using a finite random test set
- Interpreting empirical metrics without errors is dangerous

Presenting Metrics

- Measuring performance comes with errors
- There may be systematic error due to the biases in your test set—think carefully about is your test fair
- There will also be statistical errors coming from using a finite random test set
- Interpreting empirical metrics without errors is dangerous

Presenting Metrics

- Measuring performance comes with errors
- There may be systematic error due to the biases in your test set—think carefully about is your test fair
- There will also be statistical errors coming from using a finite random test set
- Interpreting empirical metrics without errors is dangerous

Presenting Metrics

- Measuring performance comes with errors
- There may be systematic error due to the biases in your test set—think carefully about is your test fair
- There will also be statistical errors coming from using a finite random test set
- Interpreting empirical metrics without errors is dangerous and unnecessary

Presenting Metrics

- Measuring performance comes with errors
- There may be systematic error due to the biases in your test set—think carefully about is your test fair
- There will also be statistical errors coming from using a finite random test set
- Interpreting empirical metrics without errors is dangerous and unnecessary
- **Lesson 12: Always compute estimated errors in the mean**

Errorbars for Bernoulli Trials

- If you have some binary measurement and you want to estimate the probability of success, p, then your estimated error in mean is

$$\Delta = \sqrt{\frac{p\,(1-p)}{n}}$$

- Of course, you don't know p, but the worst it gets is when $p = \frac{1}{2}$ (giving errors in the mean of $\Delta = \frac{1}{2\sqrt{n}}$)
- For $n = 10\,000$ this is 0.5%
- You can get a tighter bound by using $p \approx \hat{p} = s/n$
- This only fails if the number of successes, s, is very small (or zero)

Errorbars for Bernoulli Trials

- If you have some binary measurement and you want to estimate the probability of success, p, then your estimated error in mean is

$$\Delta = \sqrt{\frac{p\,(1-p)}{n}}$$

- Of course, you don't know p, but the worst it gets is when $p = \frac{1}{2}$ (giving errors in the mean of $\Delta = \frac{1}{2\sqrt{n}}$)
- For $n = 10\,000$ this is 0.5%
- You can get a tighter bound by using $p \approx \hat{p} = s/n$
- This only fails if the number of successes, s, is very small (or zero)

Errorbars for Bernoulli Trials

- If you have some binary measurement and you want to estimate the probability of success, p, then your estimated error in mean is

$$\Delta = \sqrt{\frac{p\,(1-p)}{n}}$$

- Of course, you don't know p, but the worst it gets is when $p = \frac{1}{2}$ (giving errors in the mean of $\Delta = \frac{1}{2\sqrt{n}}$)
- For $n = 10\,000$ this is 0.5%
- You can get a tighter bound by using $p \approx \hat{p} = s/n$
- This only fails if the number of successes, s, is very small (or zero)

Errorbars for Bernoulli Trials

- If you have some binary measurement and you want to estimate the probability of success, p, then your estimated error in mean is

$$\Delta = \sqrt{\frac{p\,(1-p)}{n}}$$

- Of course, you don't know p, but the worst it gets is when $p = \frac{1}{2}$ (giving errors in the mean of $\Delta = \frac{1}{2\,\sqrt{n}}$)
- For $n = 10\,000$ this is 0.5%
- You can get a tighter bound by using $p \approx \hat{p} = s/n$
- This only fails if the number of successes, s, is very small (or zero)

Errorbars for Bernoulli Trials

- If you have some binary measurement and you want to estimate the probability of success, p, then your estimated error in mean is

$$\Delta = \sqrt{\frac{p\,(1-p)}{n}}$$

- Of course, you don't know p, but the worst it gets is when $p = \frac{1}{2}$ (giving errors in the mean of $\Delta = \frac{1}{2\sqrt{n}}$)
- For $n = 10\,000$ this is 0.5%
- You can get a tighter bound by using $p \approx \hat{p} = s/n$
- This only fails if the number of successes, s, is very small (or zero)

Errorbars for Continuous Variables

- If your measurement is not binary then $\Delta = \frac{\sigma}{n}$, where σ is your standard deviation

- You usually use the empirically measured standard deviation

$$\sigma \approx \hat{\sigma} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \hat{\mu})^2}$$

- The only time this will fail is when your random variables, X_i, come from a distribution with power-law tails that don't necessarily have a variance

- Although you need $n > 10$ to ensure $\sigma \approx \hat{\sigma}$

Errorbars for Continuous Variables

- If your measurement is not binary then $\Delta = \frac{\sigma}{n}$, where σ is your standard deviation

- You usually use the empirically measured standard deviation

$$\sigma \approx \hat{\sigma} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \hat{\mu})^2}$$

- The only time this will fail is when your random variables, X_i, come from a distribution with power-law tails that don't necessarily have a variance

- Although you need $n > 10$ to ensure $\sigma \approx \hat{\sigma}$

Errorbars for Continuous Variables

- If your measurement is not binary then $\Delta = \frac{\sigma}{n}$, where σ is your standard deviation

- You usually use the empirically measured standard deviation

$$\sigma \approx \hat{\sigma} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \hat{\mu})^2}$$

- The only time this will fail is when your random variables, X_i, come from a distribution with power-law tails that don't necessarily have a variance

- Although you need $n > 10$ to ensure $\sigma \approx \hat{\sigma}$

Errorbars for Continuous Variables

- If your measurement is not binary then $\Delta = \frac{\sigma}{n}$, where σ is your standard deviation

- You usually use the empirically measured standard deviation

$$\sigma \approx \hat{\sigma} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \hat{\mu})^2}$$

- The only time this will fail is when your random variables, X_i, come from a distribution with power-law tails that don't necessarily have a variance

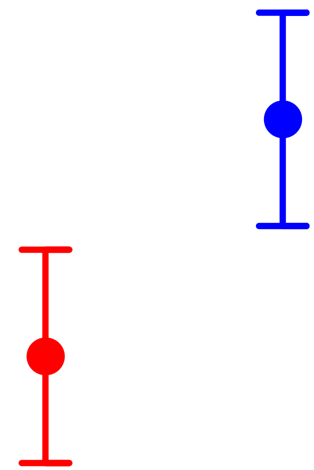
- Although you need $n > 10$ to ensure $\sigma \approx \hat{\sigma}$

Errors and Statistical Significance

- Errors in the mean are necessary to tell if two number are statistically significantly different

- A rule of thumb is they will be statistically significantly different if their errorbars don't intercept

- You can, of course, use t-tests to verify this

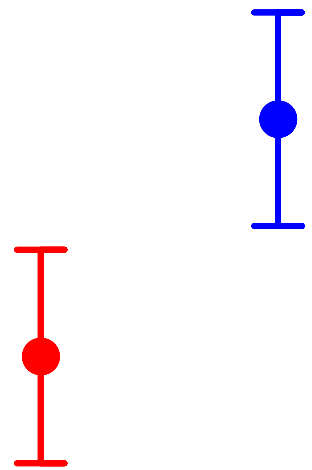


Errors and Statistical Significance

- Errors in the mean are necessary to tell if two number are statistically significantly different

- A rule of thumb is they will be statistically significantly different if their errorbars don't intercept

- You can, of course, use t-tests to verify this

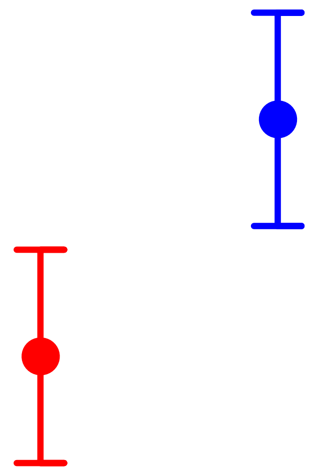


Errors and Statistical Significance

- Errors in the mean are necessary to tell if two number are statistically significantly different

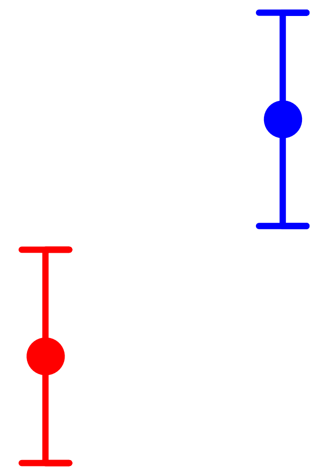
- A rule of thumb is they will be statistically significantly different if their errorbars don't intercept

- You can, of course, use t-tests to verify this



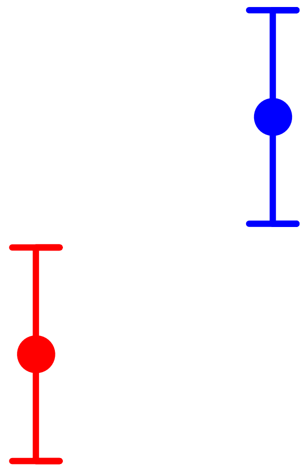
Errors and Statistical Significance

- Errors in the mean are necessary to tell if two number are statistically significantly different
- A rule of thumb is they will be statistically significantly different if their errorbars don't intercept
- You can, of course, use t-tests to verify this
- **Lesson 13: Compute statistical significance when necessary**



Errors and Statistical Significance

- Errors in the mean are necessary to tell if two number are statistically significantly different
- A rule of thumb is they will be statistically significantly different if their errorbars don't intercept
- You can, of course, use t-tests to verify this
- **Lesson 13: Compute statistical significance when necessary**
- **Lesson 14: Statistical significance doesn't necessarily mean this is a difference you care about**



Standard Deviation and Errors In Mean

- You sometimes care about how a metric varies between trials
- This is commonly measured using standard deviation
- Although it is common to use box plot, violin plots to show this variation graphically
- These don't show your uncertainty in the mean value, only the sample-to-sample variability

Standard Deviation and Errors In Mean

- You sometimes care about how a metric varies between trials
- This is commonly measured using standard deviation
- Although it is common to use box plot, violin plots to show this variation graphically
- These don't show your uncertainty in the mean value, only the sample-to-sample variability

Standard Deviation and Errors In Mean

- You sometimes care about how a metric varies between trials
- This is commonly measured using standard deviation
- Although it is common to use box plot, violin plots to show this variation graphically
- These don't show your uncertainty in the mean value, only the sample-to-sample variability

Standard Deviation and Errors In Mean

- You sometimes care about how a metric varies between trials
- This is commonly measured using standard deviation
- Although it is common to use box plot, violin plots to show this variation graphically
- These don't show your uncertainty in the mean value, only the sample-to-sample variability

Empirically Computing Statistical Significance

- You often want to show your result is better than chance
- You can compute a p-value by simulating the chance event and seeing what proportion of the time you exceed your result
- If you don't exceed your result in a million trials your p-value is less than 10^{-6} (i.e. super significant)
- Often doing these tests allow you to make a strong statement

Empirically Computing Statistical Significance

- You often want to show your result is better than chance
- You can compute a p-value by simulating the chance event and seeing what proportion of the time you exceed your result
- If you don't exceed your result in a million trials your p-value is less than 10^{-6} (i.e. super significant)
- Often doing these tests allow you to make a strong statement

Empirically Computing Statistical Significance

- You often want to show your result is better than chance
- You can compute a p-value by simulating the chance event and seeing what proportion of the time you exceed your result
- If you don't exceed your result in a million trials your p-value is less than 10^{-6} (i.e. super significant)
- Often doing these tests allow you to make a strong statement

Empirically Computing Statistical Significance

- You often want to show your result is better than chance
- You can compute a p-value by simulating the chance event and seeing what proportion of the time you exceed your result
- If you don't exceed your result in a million trials your p-value is less than 10^{-6} (i.e. super significant)
- Often doing these tests allow you to make a strong statement

Empirically Computing Statistical Significance

- You often want to show your result is better than chance
- You can compute a p-value by simulating the chance event and seeing what proportion of the time you exceed your result
- If you don't exceed your result in a million trials your p-value is less than 10^{-6} (i.e. super significant)
- Often doing these tests allow you to make a strong statement

Empirically Computing Statistical Significance

- You often want to show your result is better than chance
- You can compute a p-value by simulating the chance event and seeing what proportion of the time you exceed your result
- If you don't exceed your result in a million trials your p-value is less than 10^{-6} (i.e. super significant)
- Often doing these tests allow you to make a strong statement
- **Lesson 15: Think carefully about how to present your results convincingly**

Summary

- There are a lot of lessons to learn from past malpractice

- Following some of them (testing of different datasets, multiple models, providing code, optimising other people's code) is a pain to do

- However, many referees are going to insist you do this

- Following best practice will ultimately make you a better scientist

Summary

- There are a lot of lessons to learn from past malpractice, but they are mainly about using commonsense
- Following some of them (testing of different datasets, multiple models, providing code, optimising other people's code) is a pain to do
- However, many referees are going to insist you do this
- Following best practice will ultimately make you a better scientist

Summary

- There are a lot of lessons to learn from past malpractice, but they are mainly about using commonsense
- Following some of them (testing of different datasets, multiple models, providing code, optimising other people's code) is a pain to do
- However, many referees are going to insist you do this
- Following best practice will ultimately make you a better scientist

Summary

- There are a lot of lessons to learn from past malpractice, but they are mainly about using commonsense
- Following some of them (testing of different datasets, multiple models, providing code, optimising other people's code) is a pain to do
- However, many referees are going to insist you do this
- Following best practice will ultimately make you a better scientist

Summary

- There are a lot of lessons to learn from past malpractice, but they are mainly about using commonsense
- Following some of them (testing of different datasets, multiple models, providing code, optimising other people's code) is a pain to do
- However, many referees are going to insist you do this
- Following best practice will ultimately make you a better scientist

Summary

- There are a lot of lessons to learn from past malpractice, but they are mainly about using commonsense
- Following some of them (testing of different datasets, multiple models, providing code, optimising other people's code) is a pain to do
- However, many referees are going to insist you do this
- Following best practice will ultimately make you a better scientist
- **Lesson 16: Using robust commonsense can go a long way to avoid common errors**